



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Augmented Reality Software for Interior Design

A Software Engineering Project Submitted

By

Semester: Fall 23-24		Section: H	Group Number: 5	
SN	Student Name	Student ID	Contribution (CO1+CO2)	Individual Marks
1.	Md. Redowan Hossain	21-45445-3	Equally contributed in CO1 & CO2.	
2.	Md. Amir Hossain Alif	21-45446-3	Equally contributed in CO1 & CO2.	
3.	Raiyan Sayeed	22-46494-1	Equally contributed in CO1 & CO2.	
4.	Shakil Jamil Mrida	22-47275-1	Equally contributed in CO1 & CO2.	
5.	Nowrin Tasnim	22-47280-1	Equally contributed in CO1 & CO2.	

The project will be evaluated for the following Course Outcomes

CO1: <i>Analyze</i> the impact of software engineering models over various context of software development to assess societal, health, safety, legal and cultural issues.	Total Marks	
	Project Background Analysis and feasibility (needs, goal, benefits, etc.)	[5 Marks]
	Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]
CO2: <i>Explain</i> appropriate software engineering model, project management roles and their skills in the context of professional engineering practice and solutions to complex engineering problems in a software development environment.	Review of existing Studies and Relevant Example	[5Marks]
	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5Marks]
	Evidence of Argumentation regarding process model selection	[5Marks]
	Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report	[5Marks]

Description of Student's Contribution in the Project work

Student Name: Md. Redowan Hossain

Student ID: 21-45445-3

Contribution in Percentage (20%):

Contribution in the Project:

- Project Proposal,
- Software Project Model Selection,
- Specification Document (Design, Side Navigation)
- Class Diagram.
- UI/UX Design (MyDownloads, Most used & Favorite Templates, Work History)
- Project Testing (Design, Side Navigation)
- WBS Diagram
- Timeline Chart 1 & 2
- Risk Analysis

Md. Redowan Hossain

Signature of the Student

Student Name: Md. Amir Hossain Alif

Student ID: 21-45446-3

Contribution in Percentage (20%):

Contribution in the Project:

- Project Proposal,
- Software Project Model Selection,
- Specification Document (Search Bar, Notification)
- Use case Diagram
- UI/UX Design (Login, SignUp, Forgot Password)
- Project Testing (Search Bar, Notification)
- WBS diagram
- Timeline Chart 1 & 2
- Risk Analysis

Md. Amir Hossain Alif

Signature of the Student

Student Name: Raiyan Sayeed

Student ID: 22-46494-1

Contribution in Percentage (20%):

Contribution in the Project:

- Project Proposal,
- Software Project Model Selection,
- Specification Document (Home Page, Subscription) & Non-functional requirements

- Activity Diagram.
- UI/UX Design (Homepage, Design)
- Project Testing (Subscription)
- WBS Diagram
- Timeline Chart 1, 2 & EVA calculations.
- Risk Analysis

Raiyan Sayeed

Signature of the Student

Student Name: Shakil Jamil Mrida

Student ID: 22-47275-1

Contribution in Percentage (20%):

Contribution in the Project:

- Project Proposal,
- Software Project Model Selection,
- Specification Document (Login, SignUp, Forgot Password)
- Sequence Diagram.
- UI/UX Design (Side Navigation, Subscription).
- Project Testing (Login, SignUp, Forgot Password)
- WBS Diagram and CoCoMo calculations.
- Timeline Chart 1
- Risk Analysis

Shakil Jamil Mrida

Signature of the Student

Student Name: Nowrin Tasnim

Student ID: 22-47280-1

Contribution in Percentage (20%):

Contribution in the Project:

- Project Proposal,
- Software Project Model Selection,
- Specification Document (MyDownloads, Most Used & Favorite Templates, Work History)
- UI/UX Design (Search Bar, Notifications).
- Project Testing (MyDownloads, Most Used & Favorite Templates, Work History)
- Timeline Chart 2
- Risk Analysis

Nowrin Tasnim

Signature of the Student

1. PROJECT PROPOSAL

Background:

The process of interior design is a fairly complicated process, and it involves multifarious colours, furniture, fittings and layout selection. Traditionally, this process has been shown using drawings, 2D and 3D rendering software, but these approaches are quite time-consuming and may not give us or the clients their desired output. Additionally, clients often cannot visualize or understand how the proposed design would look in their space/place/room which may lead to dissatisfaction and costly re-modifications.

We believe that the fundamental source of this problem is the difference or gap between the abstract representation of the designs and the actual reality of the spaces. As a result, this may cause a serious misunderstanding between designer and client which leads to higher prices and not meeting deadlines.

Solution:

The goal of our project is to provide Augmented Reality (AR) software for interior design that allows users to visualize designs in their actual physical location by the screens of their smartphones, tablets etc. The approach is especially relevant since AR technology may bridge the gap between abstract design representations and actual reality, resulting in a more intuitive and satisfying design experience. This approach is in line with current industry trends toward digitalization and immersive technology. It provides a user-friendly solution and enhances the collaboration between designers, architects, and clients, resulting in better design ideas.

Using this software, users will be able to choose their own furniture and décor items from a large database and visually put them in their homes using AR software. It will also include tools for altering the colours of walls and flooring materials. This program will enable users to make educated design decisions by employing cutting-edge AR technology, decreasing the chance of costly errors and disappointment. AR technology improves construction project safety and quality by providing real-time viewing of plans, highlighting possible dangers and errors before implementation.

Interior designers and their clients will be the key users of this program. The utilization of AR technology necessitates careful consideration of intellectual property rights and licensing for 3D models and furniture designs, as well as resolving privacy concerns to guarantee safe and compliant management of user data. Designers will profit from a more efficient design process and greater client communication, while clients will acquire a better grasp of suggested ideas, resulting in more satisfactory results. AR software in interior design may provide a competitive advantage by providing detailed information about future projects, appealing to new-age clients who seek personalization and immersive experiences, and taking cultural sensitivities into account while proposing design alternatives and furniture selections which can appeal to people of different cultures.

This software will help to advance the area of augmented reality in interior design by delivering a practical, user-friendly solution. It will expand on previous research into the design of AR apps, improving their functionality and simplicity of use.

While various augmented reality (AR) applications for interior design exist, such as IKEA Place, Houzz, Dulux Visualizer, MagicPlan, DecorMatters, and others, this software will provide more complete design capabilities and a wider product database, giving consumers more freedom and possibilities. This project will improve the area of Augmented Reality (AR) in interior design by building on the strengths of existing solutions while resolving their limitations.

2. SOFTWARE PROJECT MODEL SELECTION

As our project “Augmented Reality Software for Interior Design” is mostly iterative in nature and requirements may change over time, the Agile Software Development Model, specifically SCRUM, would be the most suitable process for it. We selected the SCRUM process model since it puts an emphasis on iterative development, flexibility and teamwork. It will allow us to continuously refine our project, adapting swiftly to changes as per the evolving requirements as the field of AR is constantly developing. As we have a small group, regular SCRUM meetings will promote good communication between the team members thereby ensuring the timely resolution

of any issue and enhancing the quality of the project/software. Last but not least in the SCRUM process, the whole project is divided into manageable sprints, which allows the team to focus on one segment of the project at one time.

The other Agile Software Development models were not suitable for our project, “Augmented Reality Software for Interior Design” for a multitude of reasons. First is Extreme Programming (XP), which, although mostly customer-focused and flexible, relies too heavily on face-to-face communication and pair programming which is not that suitable for our team. It also limits strategies as we have to follow the current user stories accordingly. Next, the Dynamic Systems Development Method (DSDM) is constrained by time and budget, which could limit flexibility when changes are needed, and it also requires a clear understanding of requirements from the beginning, which may not be feasible for our project as AR and Interior Design is continually developing. Finally, as our team is smaller in size, the Feature Driven Development (FDD) is not suitable as it is only applicable to large projects and teams. It also involves much documentation which may slow down our project development cycle. Taking these facts into account, we chose the SCRUM model over the other Agile Software development models owing to its high flexibility, small sprints, and well-structured framework for our project.

Roles and responsibilities of the SCRUM model in our project:

1. **Scrum Master:** In our project, the Scrum Master will ensure that the team follows the SCRUM process, help in communication between the members, and try to resolve any obstacle that can hinder our development progress. He will also set up the daily SCRUM meetings, and sprint planning and will also help the clients and management understand the intricate features of the app.
2. **Product Owner:** The Product Owner will be responsible for defining the product goals for our “Augmented Reality Software for Interior Design” application, prioritizing the product backlog based on user/customer needs related to interior design and business requirements, and ensuring the development team is focused on delivering value to customers and the business.
3. **Scrum Team:** The Scrum Team will be responsible for the overall development of the project and will therefore be responsible for accomplishing the tasks set by the Product

Owner for a particular sprint. Besides developing the software, they will also test and update the software based on customer feedback.

4. **Customer:** The customer, for our project, will be the architects or interior designers or the ones who will use our software. They are the ones who will provide feedback during the sprint review meetings, which will be used by the Product Owner and Scrum Team to refine their product backlog and future sprints.
5. **Management:** Management will provide necessary resources for the sprints and updates related to the AR and Interior Design to the Scrum Team. They will be involved in decision-making and deploying strategies based on customer feedback thereby deciding the direction our project will take.

3. SPECIFICATION DOCUMENT

3.1. Login:

Functional Requirements:

3.1.1. Once the user launches the software/app, it shall allow users to log in with their registered username/email and password if the “Login” option is selected. The password will be hidden.

3.1.2. The username/email and password will be verified with database records.

3.1.3. If the login is successful, the home page of the user account will be displayed.

3.1.4. If login is not successful, the user can try it again 5 times. After the 5th time, an email will be sent to the email address provided by the user as a warning or to notify the user of the unidentified login attempt. The user’s login will be blocked for 1 hour.

Priority Level: High

Precondition: User has a valid account.

Non-functional Requirements:

Availability (AV): The login system shall be available 99.9% of the time.

Performance (PE): The login system shall process login requests within 2 seconds under normal load conditions.

Usability (US): The login system shall provide clear error messages to guide the user when login fails.

3.2. SignUp:

Functional Requirements:

- 3.2.1.** The user needs to click on the “SignUp” option to go to the SignUp or Registration page if he/she does not have an account.
- 3.2.2.** The user needs to provide their full name, DOB, unique mobile number, unique email address, gender, etc.
- 3.2.3.** The user needs to write their password and confirm that password. Both of them will be hidden.
- 3.2.4.** The user needs to agree to the Terms of Conditions.
- 3.2.5.** A verification email will be sent to the email address of the new user with a link. The user shall confirm their registration by clicking on that link.
- 3.2.6.** The system shall discard the registration information if the email is not verified within 5 minutes.
- 3.2.7.** If the email address is not unique, a message box will show that there is already a registered account with that email address.

Priority Level: High

Precondition: User has a valid email address.

Non-functional Requirements:

Performance (PE): The registration system shall process registration requests within 5 seconds under normal load conditions.

Integrity (IN): The registration system shall encrypt user passwords and secure the user's personal data.

Usability (US): The registration system shall provide clear error messages to guide the user when registration fails.

Availability (AV): The registration system shall be available 99.9% of the time.

3.3. Forgot Password?:

Functional Requirements:

3.3.1. In the case, that the user forgets his/her password while trying to log in, they can click on the “Forgot Password?” option which would be available on the login page.

3.3.2. Once the user selects the "Forgot Password" option, they shall be prompted to enter their registered email address and the system will verify it with the database.

3.3.3. If the email is verified, the system/app will show the user two options for resetting their password:

3.3.3.1. By email: If the user selects this option, the system/app will send an email to the user’s registered email address with a link (the link would be active for 5 minutes) that will redirect the user to the password reset page.

3.3.3.2. By SMS: If the user selects this option, an OTP will be sent to the user’s registered mobile phone no. which will be only applicable for 1 minute. After confirming the OTP, the user will be redirected to the password reset page.

3.3.4. On the password reset page, the user shall be prompted to enter a new password and confirm the new password; which would be verified by the system/app.

3.3.5. Once it's verified, the system will update the user’s password with the new password in the database.

3.3.6. The app will notify the user that their password has been updated and redirect them to the login page so that they can log in to their account with the new password for security purposes.

Priority Level: High

Precondition: User has a valid account and access to their registered email.

Non-functional Requirements:

Availability (AV): The Forgot Password functionality should be available 99.9% of the time to ensure users can always access their accounts.

Performance (PE): The Forgot Password process should complete within 5 seconds under normal load conditions.

Integrity (IN): The system should securely handle the user's email and new password data during the Forgot Password process.

Usability (US): The Forgot Password process should be easy to understand and use for the end users.

3.4. Home Page:

Functional Requirements:

- 3.4.1.** The software/app shall display the Homepage after a successful user login.
- 3.4.2.** The Homepage shall include a **Search Bar** at the top of the page. The user can search templates and things relevant to interior design in the Search bar.
- 3.4.3.** The users can receive notifications from the **Notifications** option which will be on the right side of the Search Bar.
- 3.4.4.** The Homepage will have the “**Design**” option at the bottom of the Homepage which the user can use to capture, edit, and modify their design projects for interior design.
- 3.4.5.** The users can drag a “**Side Navigation**” bar from the side of the homepage which will allow users to access “**Profile Edit**”, “**Settings**” and “**Log out**”.
- 3.4.6.** A general user can upgrade themselves to a premium user if they select the “**Subscription**” option which will be displayed at the uppermost right corner of the homepage, to the right of the Notification option.
- 3.4.7.** The software/app shall display the following sections in the middle of the homepage: “**My Downloads**”, “**Most Used and Favorite Templates**” and “**Work History**”.
 - 3.4.7.1.** The user's downloaded designs will be displayed in the **My Downloads section**.
 - 3.4.7.2.** The user's and other users' most used and favourite templates are displayed in the “**Most Used and Favorite Templates**” section.
 - 3.4.7.3.** The user's works, designs and projects will be displayed in the **Work History** section.

Priority Level: High

Precondition: User has logged in.

Non-functional Requirements:

Availability (AV): The Home Page should be available 99.9% of the time to ensure users can always access their account and the application's main features.

Performance (PE): The Home Page should load within 2 seconds under normal load conditions.

Flexibility (FL): The Home Page should adapt to different screen sizes and orientations to provide a consistent user experience across different devices.

3.5. Search Bar:

Functional Requirements:

3.5.1. Users can enter their search queries into the Search Bar.

3.5.2. Based on the user's input, the system will provide search results displaying relevant templates and other interior design items from the database. The user can choose their desired template/item from the search results in their new project/design.

3.5.3. The Search Bar will include filters that allow users to search specifically for a type of template or item.

3.5.4. The software/app will update the search results based on the selected filters. The users will also be able to reset the filters.

3.5.5. The software/app shall provide an option for users to sort the search results based on criteria such as relevance, popularity, and date added.

3.5.6. Users will be able to see the highlighted search keyword in the search results.

Priority Level: High

Precondition: User has logged in.

Non-functional Requirements:

Performance (PE): The Search Bar should return search results within 2 seconds under normal load conditions.

Usability (US): The Search Bar should provide an intuitive interface for users to enter search queries and apply filters.

3.6. Design function:

Functional Requirements:

3.6.1. After the user selects the “**Design**” option from the homepage, first he/she needs to select “**Make a new project.**”

3.6.2. The user can select one of two options from the Design function: “**Camera Scan**” and “**Manual Modification**”.

3.6.3. Camera Scan:

3.6.3.1. After the users select this option, the system shall ask for the user's permission to access the camera the first time the "Camera Scan" option is selected.

3.6.3.2. Once the user gives permission, the software shall turn on the camera within the application.

3.6.3.3. Users can then take a photo or picture of the room or place they want to design. It will be possible to edit after the photo is taken.

3.6.3.4. Users should enter the measurements of the room so that the application properly calibrates the templates and items accordingly while placing them.

3.6.3.5. The software/app will provide the users with a variety of options for editing the photo, such as 2D and 3D templates of household items (for example, chairs, tables, dining tables, sofas, almirahs, cupboards, beds, kitchen sinks, picture frames, accessories etc.), colors, and textures which are available in the database.

3.6.3.6. The software/app will also provide templates for tiles on the floor, and designs for the walls of the room.

3.6.3.7. Users can drag, drop, and place templates on the photo of the room.

3.6.3.8. The software will automatically calibrate the size of the objects of the templates according to the room size previously inputted by the user.

3.6.3.9. For premium users, the software will provide additional templates, textures, designs and colors, and a Live Camera function. The Live Camera feature differs from the normal camera function used by general users. General users need to take multiple photos of their rooms and places at different angles and remember templates used in previous pictures to avoid overlap in the same places. The Live Camera feature allows premium users to place their templates while the live camera is ongoing, allowing them to move around their room and customize their designs. Once they are done with their design, they can stop the live camera and save their design.

3.6.3.10. Users can confirm their design by selecting "OK", which will automatically save this design in their "**Work History**".

3.6.3.11. Users can cancel their design by selecting "Cancel", which will remove all the templates placed on the photo.

3.6.3.12. The users can also choose to download their project/design which will be saved in both “**My Downloads**” and “**Work History.**”

3.6.4. Manual Modification:

3.6.4.1. This mode is especially beneficial if the user is an architect or a professional interior designer.

3.6.4.2. After the users select this option, the system shall ask for the user's permission to access the camera, gallery, and files the first time the "Manual Modification" option is selected.

3.6.4.3. Users can take a photo using the camera inside the app or upload a photo from their gallery.

3.6.4.4. The user is then redirected to a "Drawing Board" where they can make their own custom templates, designs, textures, and colours which are not available in the database of the software. The users can use a stylus or a digital/smart pen to draw as well.

3.6.4.5. After that, the software/app will render the user's drawings into 2D or 3D templates according to the user's preference.

3.6.4.6. Similar to the camera scan function, after the templates are placed on the photo, the system will automatically calibrate the size of the objects of the templates according to the room size previously inputted by the user.

3.6.4.7. The software/app will also provide an option for users to upload a custom template from their device.

3.6.4.8. Premium users can utilize additional drawing features and designs and also the Live Camera option which was mentioned previously.

3.6.4.9. Users can confirm their design by selecting "OK", which will automatically save this design in their "**Work History**".

3.6.4.10. Users can cancel their design by selecting "Cancel", which will remove all the templates placed on the photo.

3.6.4.11. The users can also choose to download their project/design which will be saved in both “**My Downloads**” and “**Work History.**”

Priority Level: High

Precondition: User is logged in.

Non-functional Requirements:

Performance (PE): The design functions should operate smoothly, with camera access and photo capture occurring within 2 seconds, and rendering of AR objects in real-time.

Usability (US): The design interface should be intuitive, allowing users to easily manipulate and place objects within their environment, with clear instructions for entering room measurements and using AR features.

Reliability (RE): The design functions should consistently produce accurate AR renderings of interior spaces, without crashes or errors.

Efficiency (EF): The application should efficiently use device resources, ensuring that AR functionalities do not drain the battery excessively or require unreasonable processing power.

3.7. Side Navigation:

Functional Requirements:

3.7.1. The software shall provide a side navigation bar for users to navigate through the application.

3.7.2. The side navigation bar shall include the following options: Profile Edit, Settings, and Logout.

3.7.3. If the users select the “**Profile Edit**” option, they will have access to the profile editing page:

3.7.3.1. The profile editing page shall allow users to update their personal information.

3.7.3.2. The profile editing page shall provide an option for users to change their registered email address and change their password.

3.7.3.3. Change Email address:

3.7.3.3.1. When the 'Change Email' option is selected, the software shall prompt the user to enter their current email address and new email address.

3.7.3.3.2. A confirmation email will be sent to the newly entered email address with a link (which will be active for 5 minutes) that confirms the new email address.

3.7.3.3.3. After confirmation, the software/app will update the new email address in the database.

3.7.3.3.4. The app will notify the user that their email address has been updated and redirect them to the login page so that they can log in to their account with the new email address for security purposes.

3.7.3.4. Change Password:

3.7.3.4.1. When the 'Change Password' option is selected, the user will be redirected to the Change Password page.

3.7.3.4.2. On the Change Password page, the user shall be prompted to enter their current password, and new password and confirm the new password; which will be verified by the system/app.

3.7.3.4.3. Once it's verified, the system will update the user's password with the new password in the database.

3.7.3.4.4. The app will notify the user that their password has been updated and redirect them to the login page so that they can log in to their account with the new password for security purposes.

3.7.3.5. All the changes made in the user's profile will be updated in the database.

3.7.4. The software/app shall allow users to access the settings page when the “**Settings**” option is selected.

3.7.4.1. The settings page shall allow users to customize their application experience.

3.7.4.2. Users can change the theme of the app in the settings.

3.7.4.3. Users can also turn on or off push notifications within the "Settings".

3.7.4.4. The software shall apply the changes made in the "Settings" immediately across the application.

3.7.4.5. The software shall provide an option to reset the settings to the default values.

3.7.5. The user shall log out of the software/app when the “**Log Out**” option is selected.

3.7.5.1. The software/app shall end the user's session when the 'Logout' option is selected.

3.7.5.2. After the user is logged out, the software shall redirect the user to the login page.

Priority Level: High

Precondition: User is logged in.

Non-functional Requirements:

Performance (PE): The Side Navigation should respond quickly to user interactions, with each option loading within 2 seconds.

Flexibility (FL): The Side Navigation should adapt to different screen sizes and orientations to provide a consistent user experience across different devices.

3.8. Notification:

Functional Requirements:

3.8.1. The software/app will provide the users with a “Notification” which will alert users of important events/updates.

3.8.2. It will generate notifications on specific triggers, for example, new templates being added to the catalogue, application updates etc.

3.8.3. It will also notify users of topics/templates according to their previous work or search preferences.

3.8.4. The users can allow these notifications to be also shown as Push Notifications on their devices which they can turn on or off in their settings.

3.8.5. If the users click on any notification, it will give them a detailed view of that notification.

3.8.6. The software/app will mark the notification as “Read” once it is viewed by the user.

Priority Level: Medium

Precondition: User is logged in and connected to the internet.

Non-functional Requirements:

Performance (PE): The Notification system should generate notifications and deliver them to the user's device within 2 seconds under normal load conditions.

Usability (US): The Notification system should provide clear and concise notifications, with an easy way for users to mark them as "Read."

Reliability (RE): The Notification system should consistently deliver notifications to users without any errors or delays.

3.9. Subscription option:

Functional Requirements:

3.9.1. The software shall provide a “**Subscription**” option for the users if they want to access the additional features of the app.

3.9.2. Once the user clicks on the “**Subscription**” option, it will redirect the user to a page that will clearly explain the features which will only be exclusive to the premium users. For example, extra 2D and 3D templates will be available for the premium users. This will be available within the application so that a general user can upgrade to a premium user if they want to.

3.9.3. If the user wants to upgrade to a premium user, they will click on the “**Confirm**” option which will redirect them to a payment gateway. The app will process the payment of the user securely and give confirmation after the completion of the successful payment. The system will update the status of the user from “general” to “premium” in the database after the payment.

3.9.4. Once the payment is successful, the software will immediately give access to the premium features to the users.

3.9.5. The software/app will allow the users to use these premium features as long as their subscription is active.

3.9.6. The software/app will also provide an option for the users to cancel their subscriptions.

3.9.7. If the user chooses to cancel their subscription, the software/app will terminate the premium access of the user at the end of their current billing period. The system will again change the status of the user from “premium” to “general” in the database after the cancellation of the subscription.

Priority Level: Medium

Precondition: User is logged in.

Non-functional Requirements:

Performance (PE): The Subscription system should process subscription requests and payment transactions within 5 seconds under normal load conditions.

Security (IN): The Subscription system should securely handle the user's payment data during the subscription process.

3.10. My Downloads:

Functional Requirements:

3.10.1. When the user selects the “My Downloads” option on the homepage, the system shall display only the projects/designs that the user has downloaded.

3.10.2. Users can again edit their downloaded designs/projects if they want to.

3.10.3. Users can also delete their downloaded designs/projects from “My Downloads” if they want to. Before deletion of a design/project, the system asks the user for reconfirmation.

3.10.4. The software/app will provide an option for users to share their downloaded designs/projects via email or social media platforms. The software shall ask for the user's permission before it shares their downloaded designs/projects.

Priority Level: High.

Precondition: User is logged in and has downloaded designs/projects.

Non-functional Requirements:

Performance (PE): The My Downloads system should load and display the user's downloaded projects/designs within 2 seconds under normal load conditions.

Usability (US): The My Downloads system should provide an intuitive interface for users to edit, delete, and share their downloaded projects/designs.

Integrity (IN): The My Downloads system should securely handle the user's downloaded projects/designs and ensure that they can only be accessed by the user.

3.11. Most Used and Favourite Templates:

Functional Requirements:

3.11.1. If the user selects the “Most Used and Favourite Templates” option on the homepage, the software/app shall display the templates that are frequently used by the user and other users of the application.

3.11.2. Users can use these templates for their new projects/designs.

3.11.3. The software/app will update the "Most Used and Favorite Templates" section based on the user's usage and the overall usage of the templates by all users.

3.11.4. The software/app will also provide an option for users to mark a template as a favorite as well, which will then be displayed in this section. The users can also remove a template from their favorites.

Priority Level: High

Precondition: User is logged in.

Non-functional Requirements:

Performance (PE): The system should display the "Most Used and Favourite Templates" within 2 seconds under normal load conditions.

Usability (US): The user interface should allow easy understanding of how to use a template and how to mark a template as a favourite.

Availability (AV): The "Most Used and Favourite Templates" feature should be available 99.9% of the time to ensure users can always access popular and favourite templates.

3.12. Work History:

Functional Requirements:

3.12.1. When the users select the "Work History" option from the homepage, the software/app will display the list of all of the projects/designs that the user has done from the first to the latest.

3.12.2. The system serializes the projects/designs according to their dates and also allows users to view the details of each project (like, size, time required to complete each project/design etc.).

3.12.3. Users can open and also edit the projects/designs from their Work History.

3.12.4. The software/app will provide an option for users to share their designs/projects via email or social media platforms. The software shall ask for the user's permission before it shares their designs/projects.

Priority Level: High

Precondition: User is logged in and has completed projects/designs.

Non-functional Requirements:

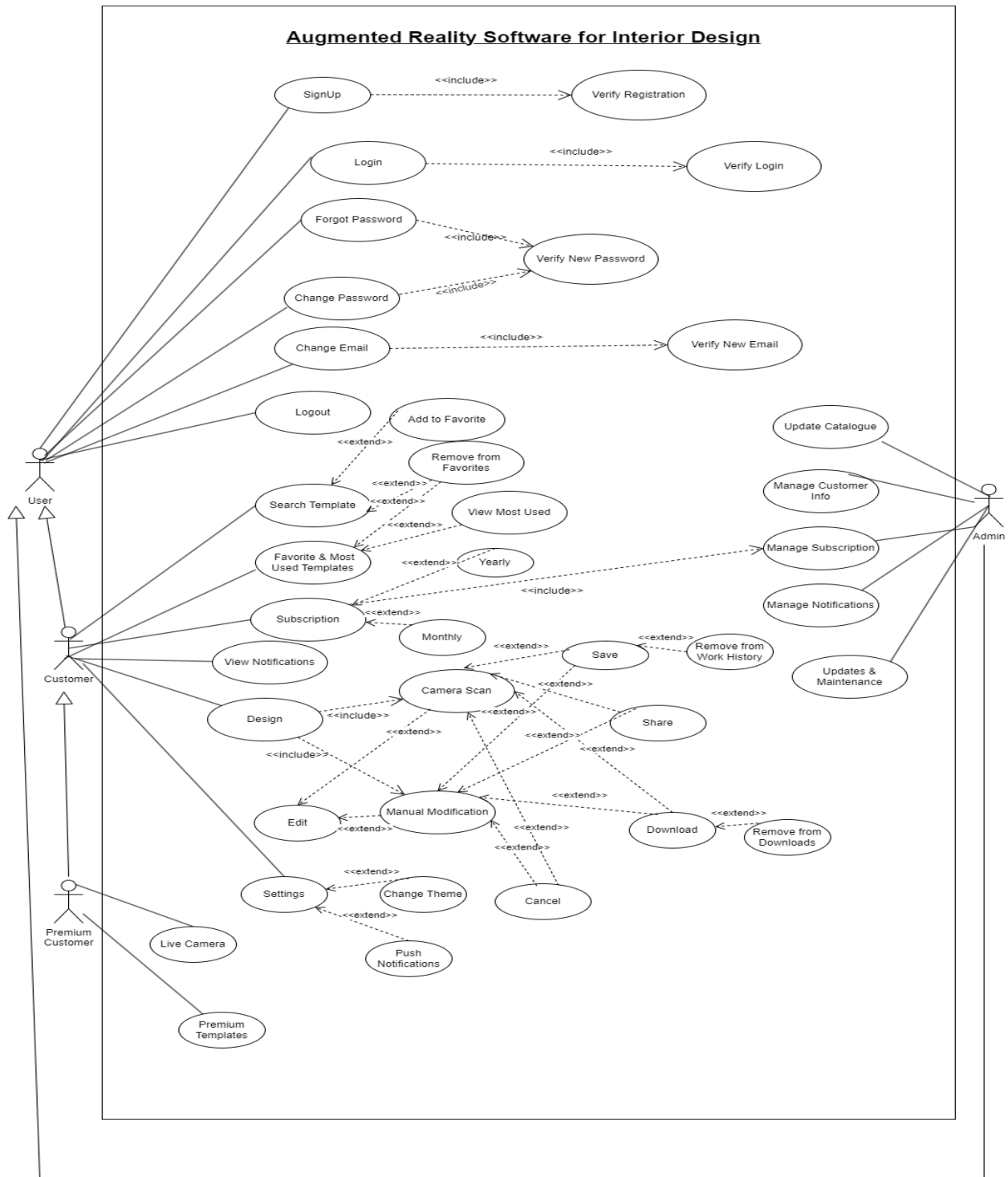
Performance (PE): The Work History system should load and display the user's work history within 2 seconds under normal load conditions.

Usability (US): The Work History system should provide an intuitive interface for users to view, edit, and share their past projects/designs.

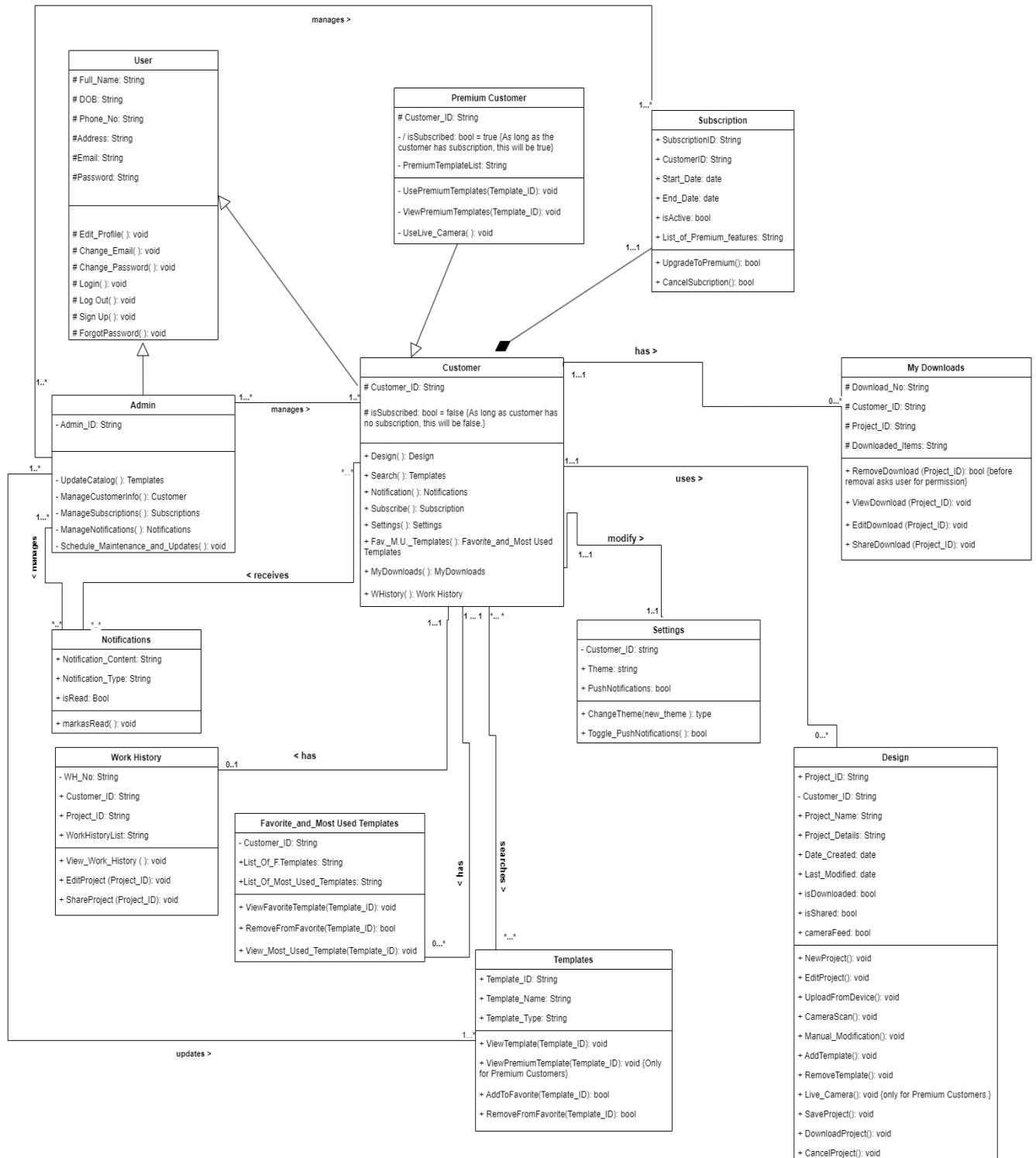
Integrity (IN): The Work History system should securely handle the user's work history and ensure that it can only be accessed by the user.

4. SYSTEM DESIGN SPECIFICATION:

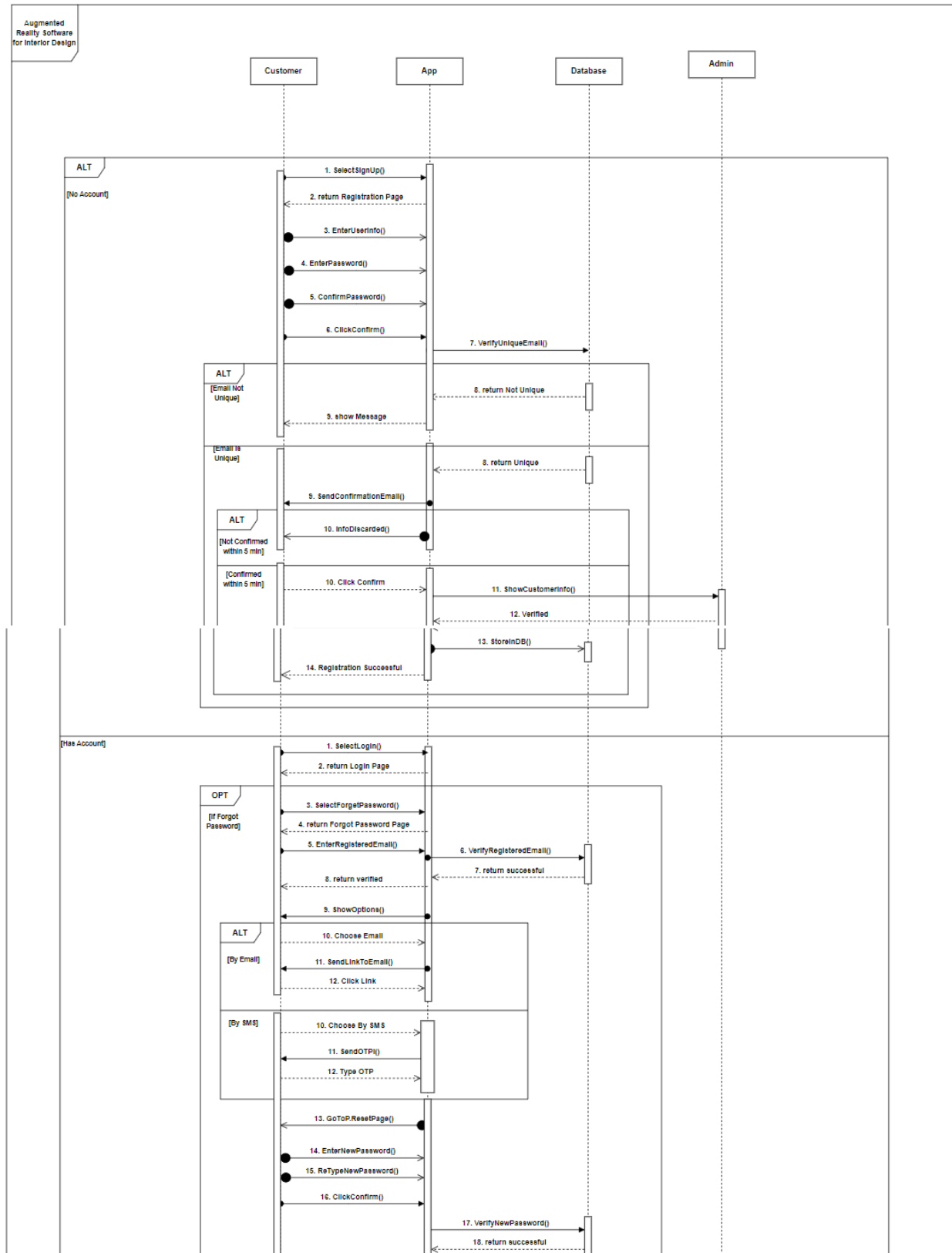
4.1. Use Case Diagram:

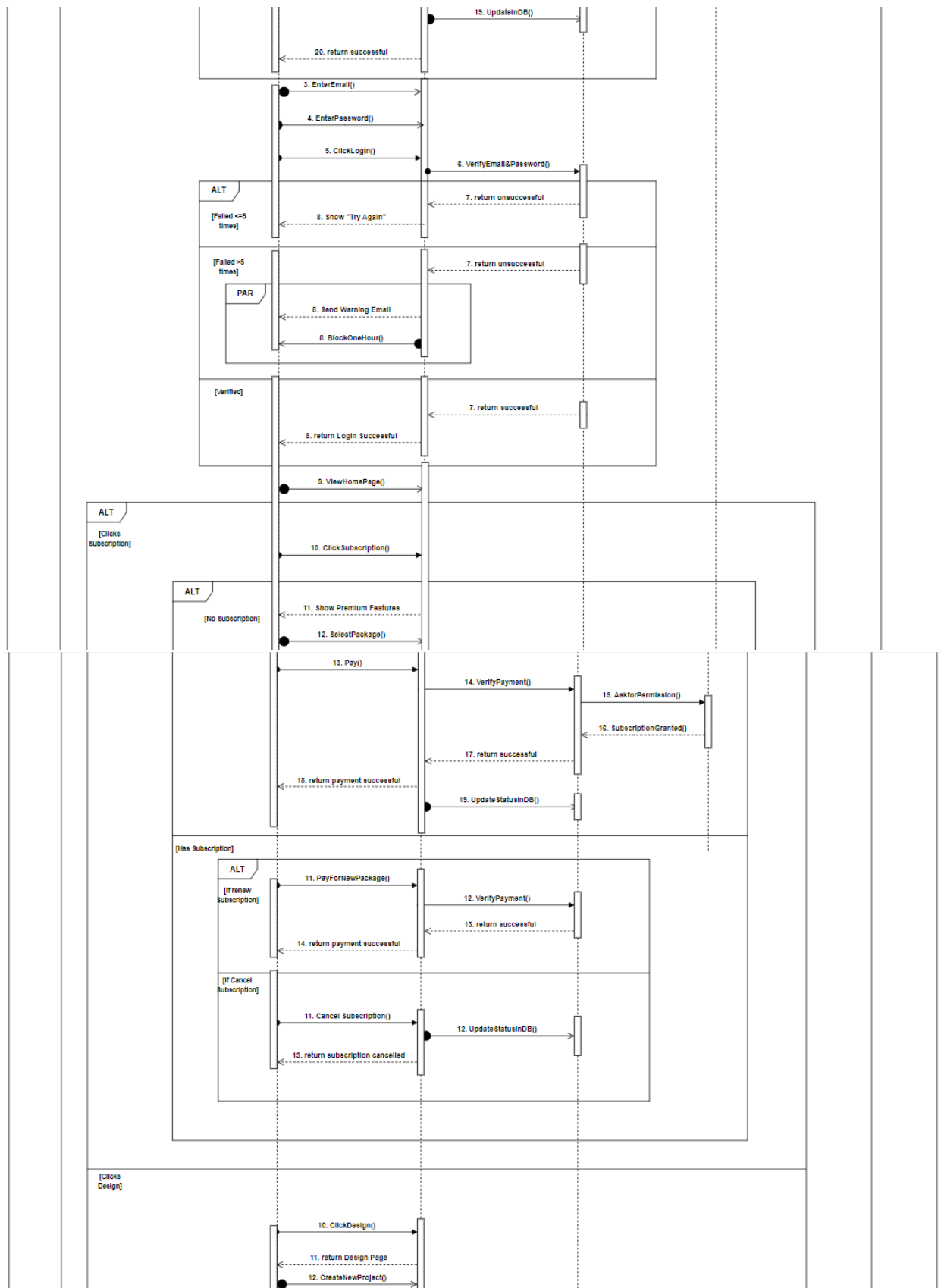


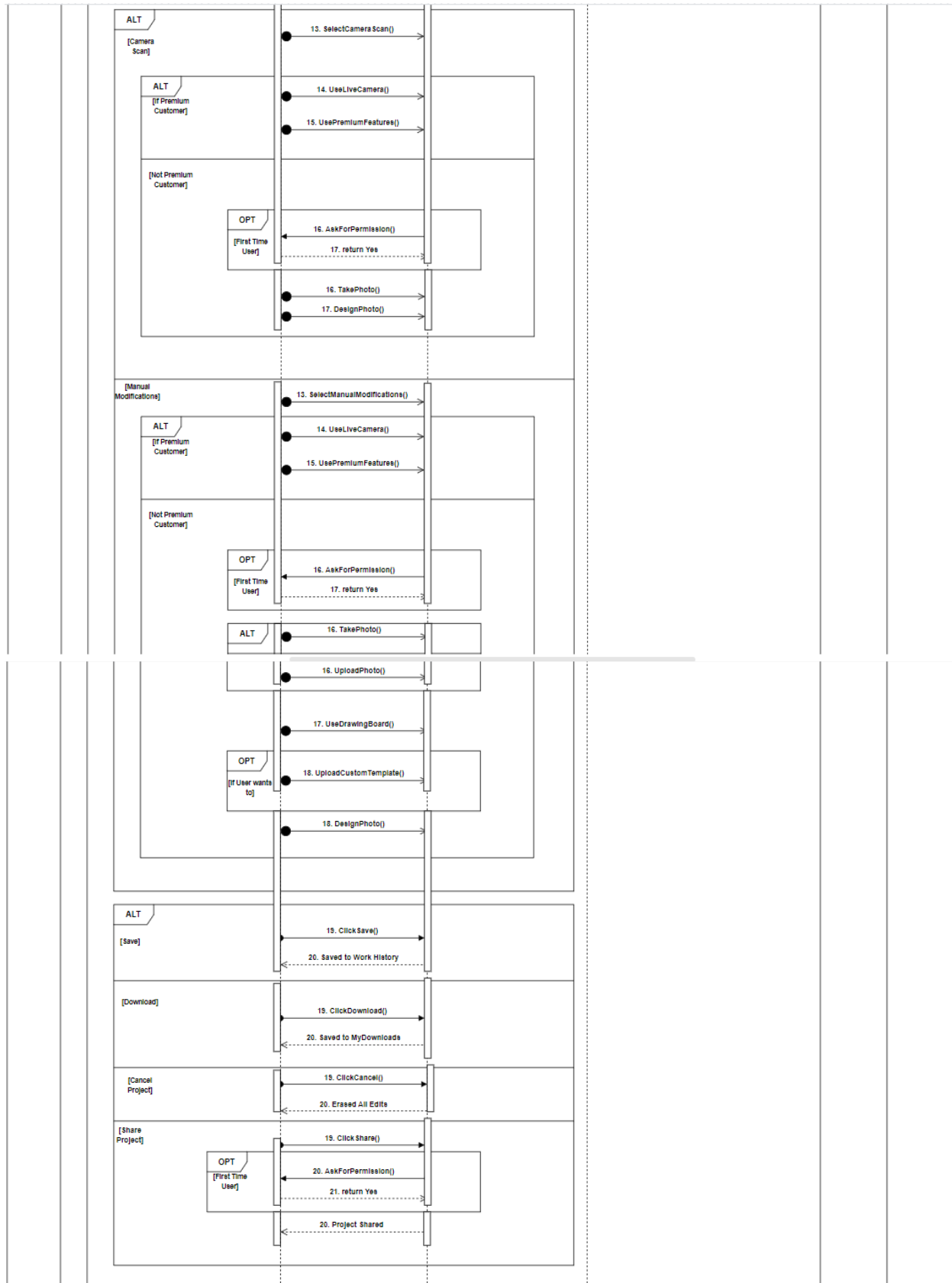
4.2. Class Diagram:

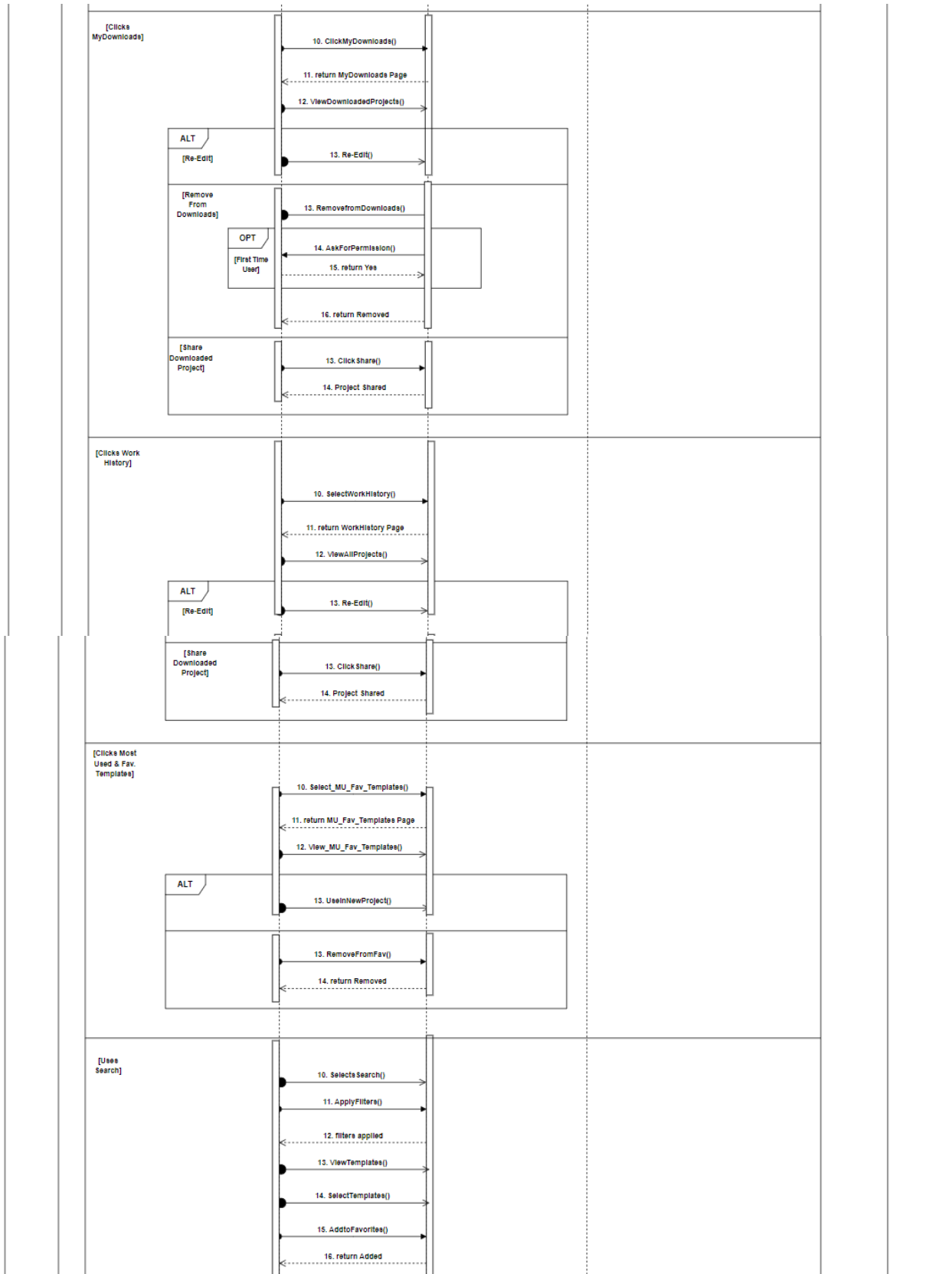


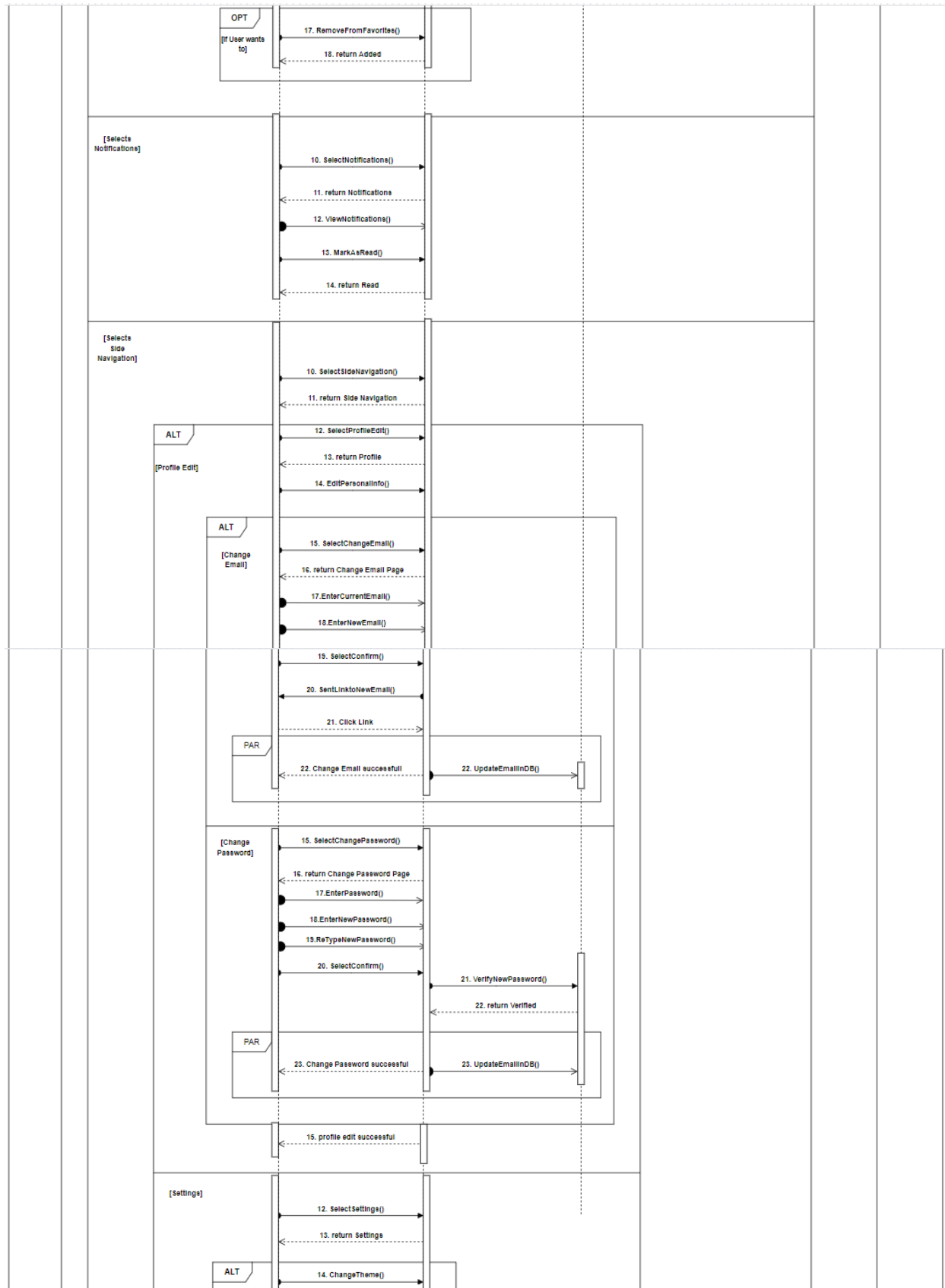
4.3. Sequence Diagram:

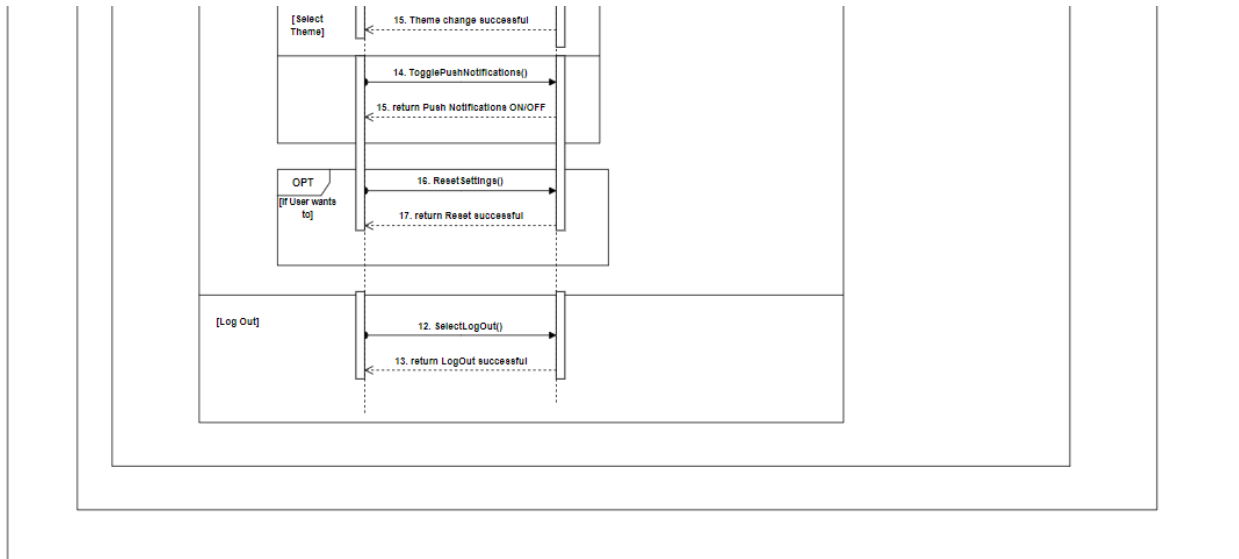




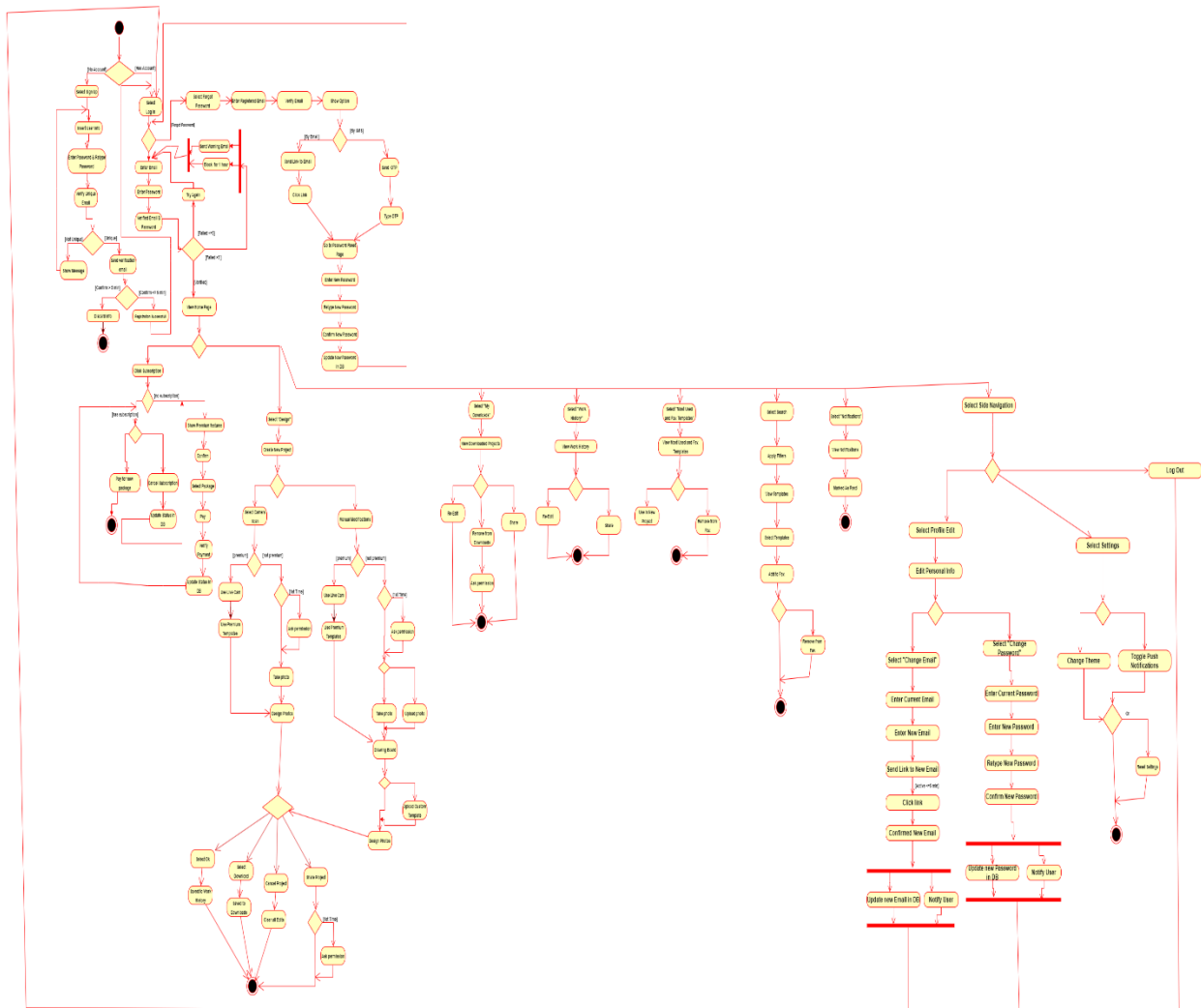






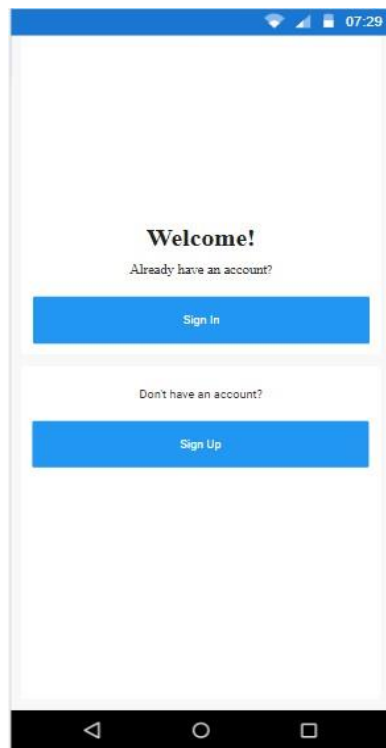


4.4. Activity Diagram:

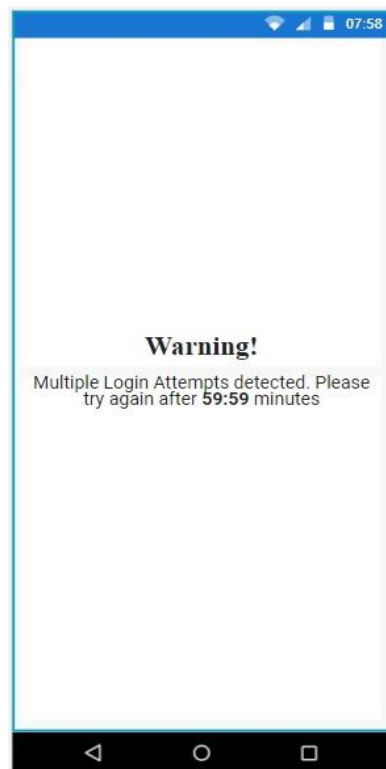


5. UI/UX DESIGN:

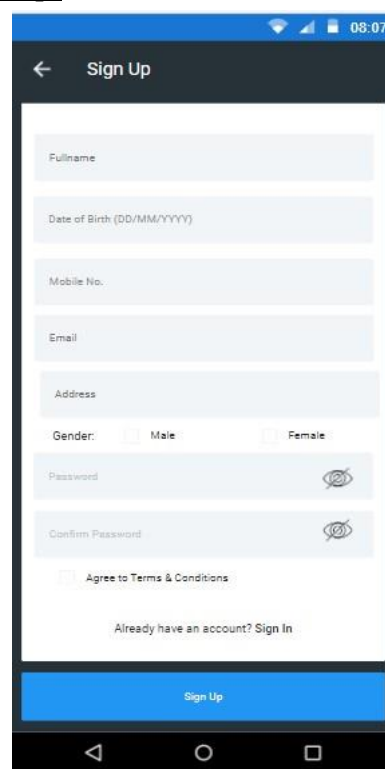
5.1. Login:

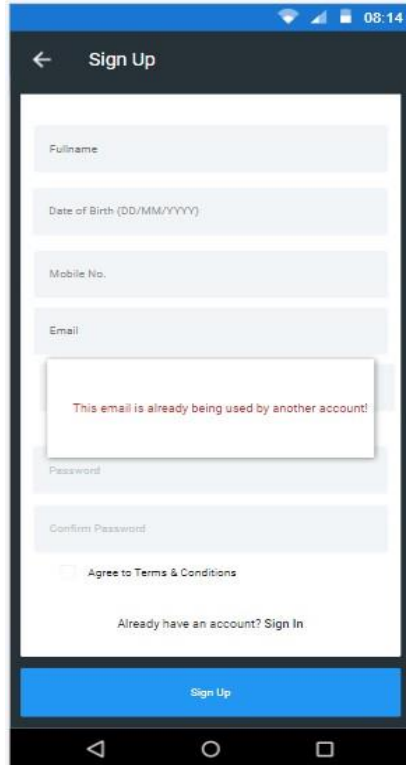


If more than 5 times:



5.2. SignUp:



5.3. Forgot Password:


Sign Up

Fullname

Date of Birth (DD/MM/YYYY)

Mobile No.

Email

This email is already being used by another account!

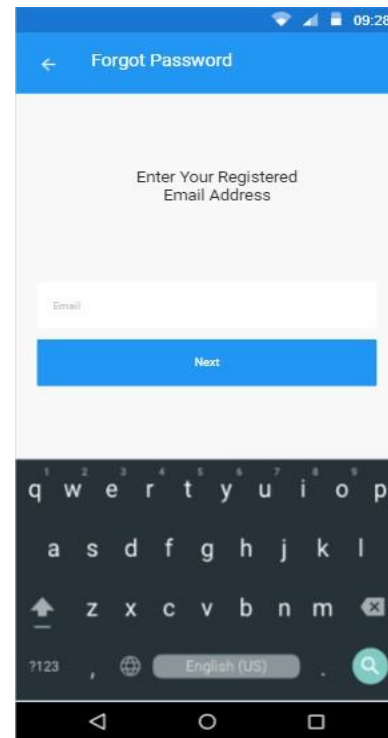
Password

Confirm Password

☐ Agree to Terms & Conditions

Already have an account? Sign In

Sign Up



Forgot Password

Enter Your Registered Email Address

Email

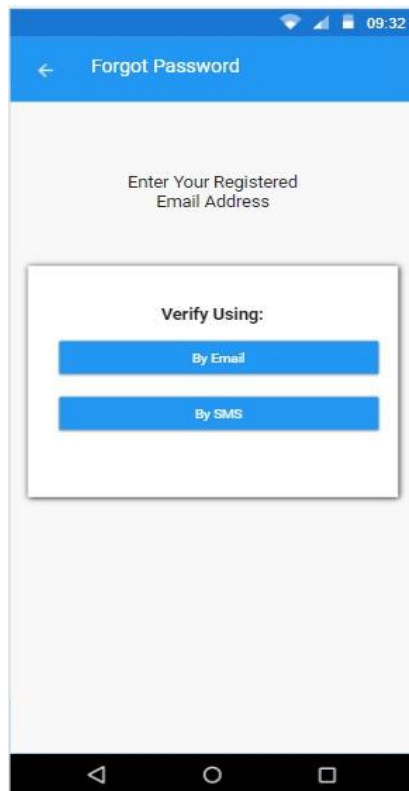
Next

q w e r t y u i o p

a s d f g h j k l

z x c v b n m

English (US)



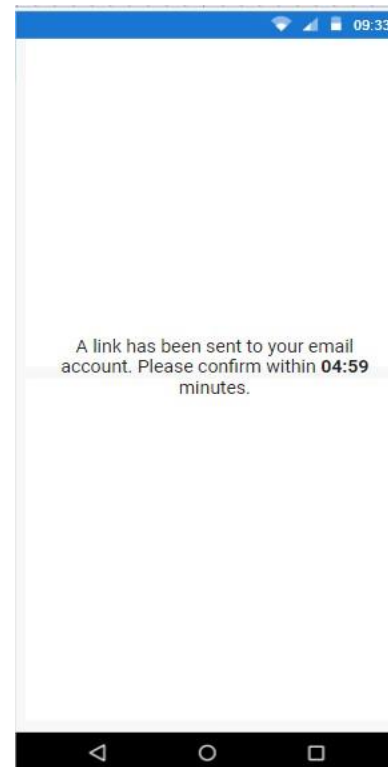
Forgot Password

Enter Your Registered Email Address

Verify Using:

By Email

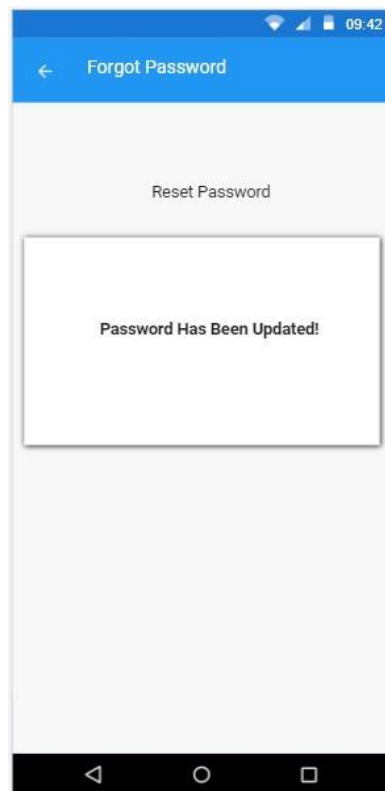
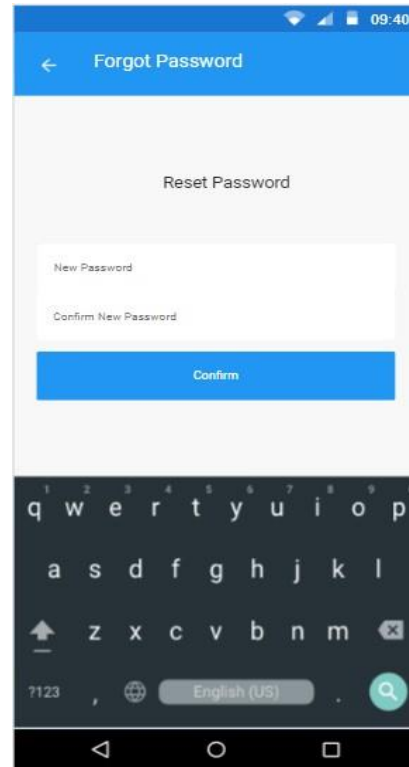
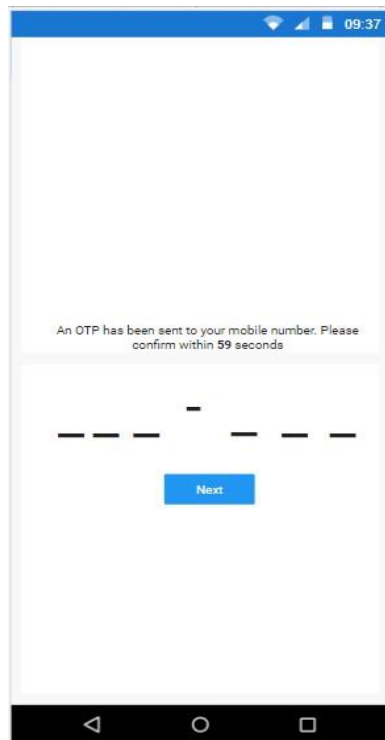
By SMS

By email:


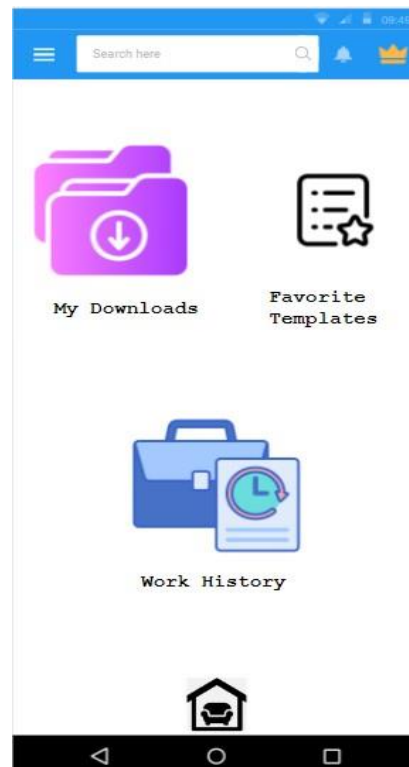
09:33

A link has been sent to your email account. Please confirm within **04:59** minutes.

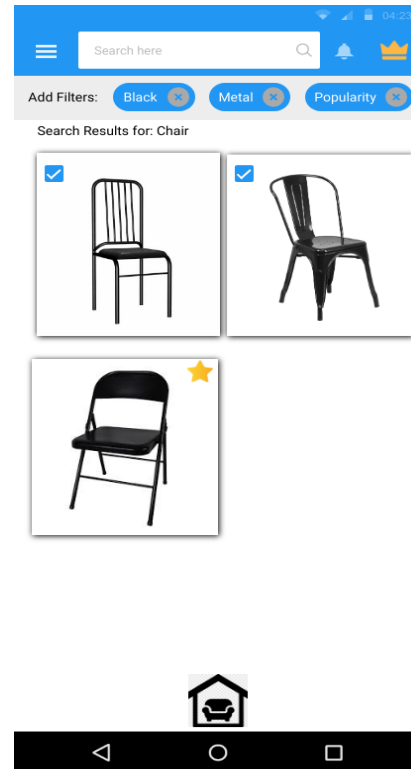
By SMS:



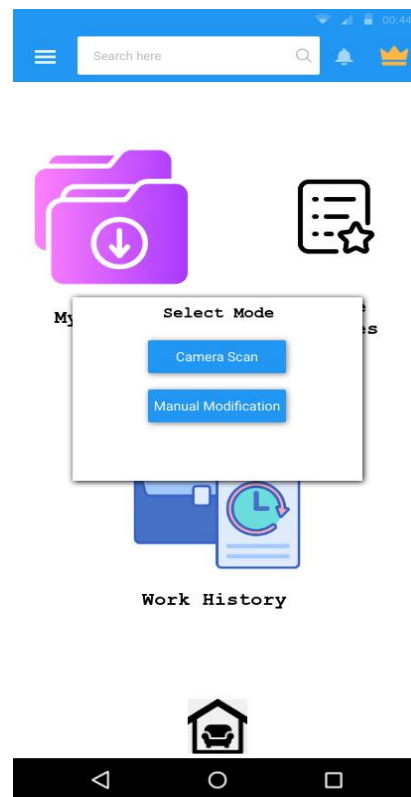
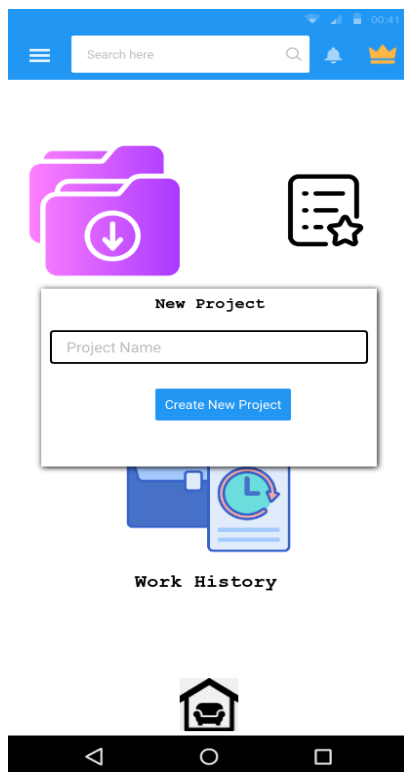
5.4. Homepage:



5.5. Search Bar:



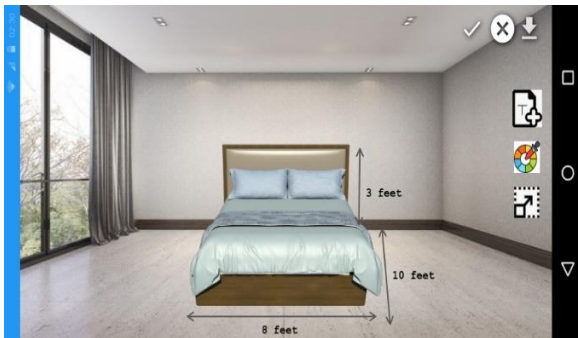
5.6. Design:

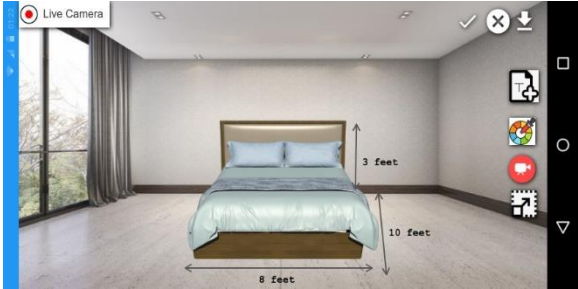
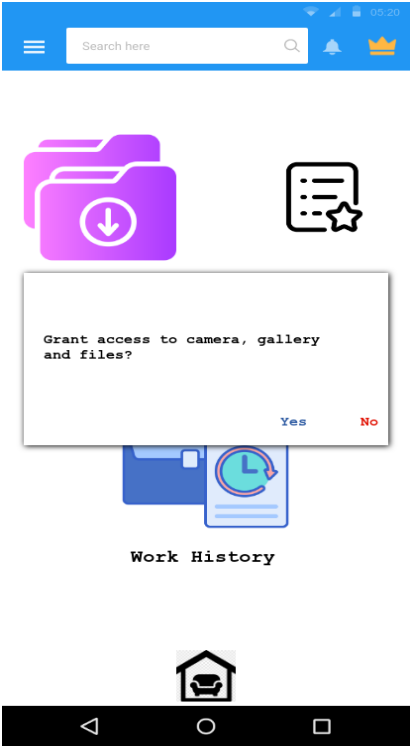
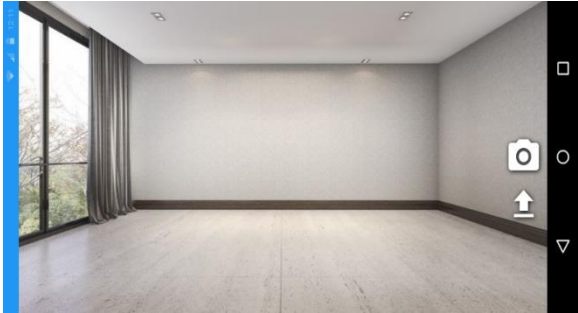


Camera Scan:

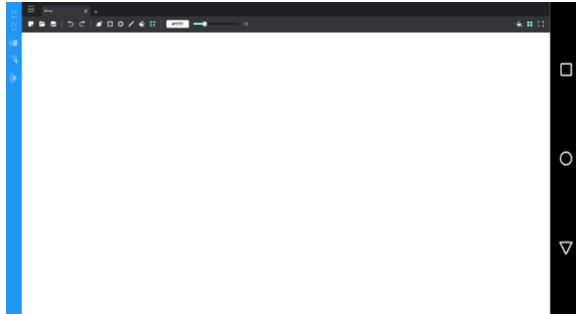


For premium users:

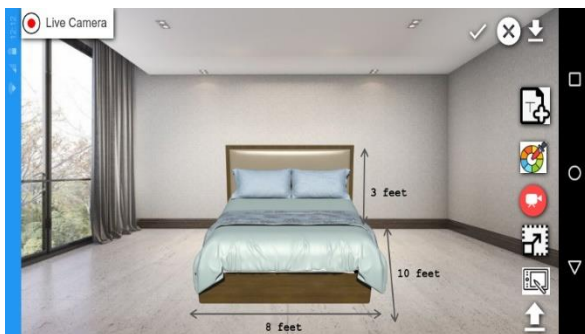
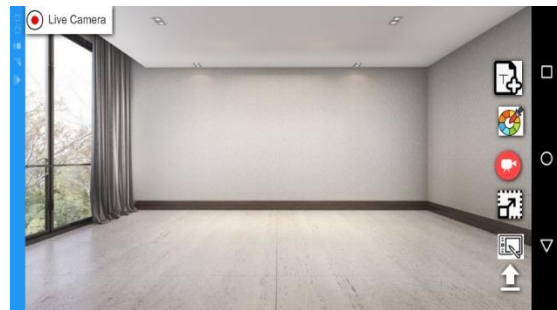


	Manual Modification: 
	

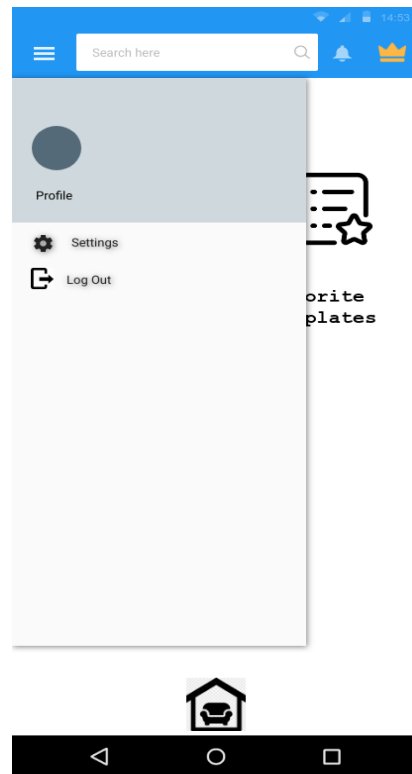
Drawing board:

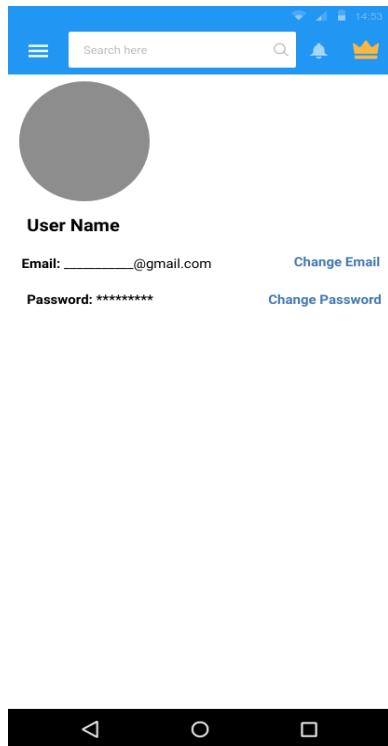


For premium users:

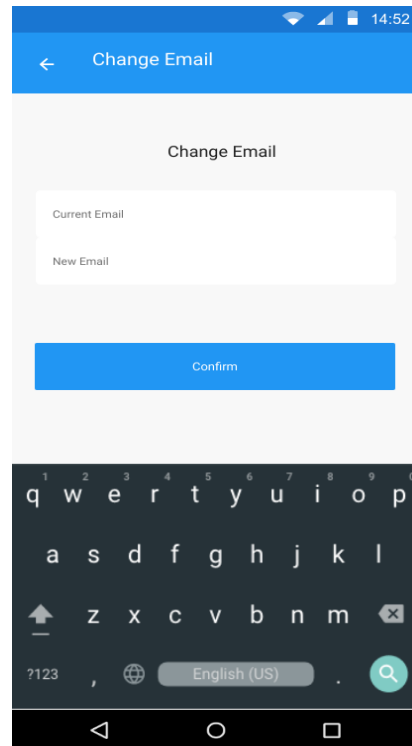


5.7. Side Navigation:

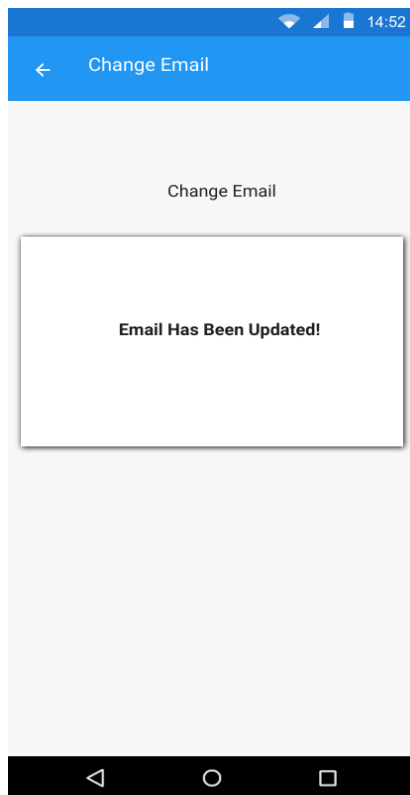


Profile:

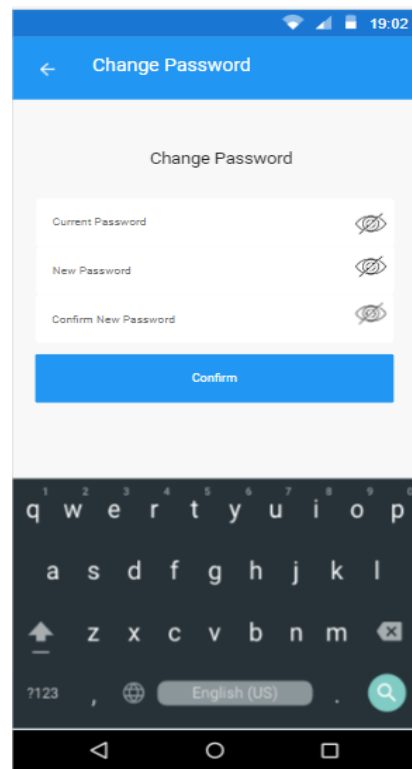
Mobile app profile screen. The header is blue with a search bar, a bell icon, and a crown icon. The status bar shows 14:53. Below the header is a grey circular profile picture. The text "User Name" is displayed. Below it, "Email: _____@gmail.com" and "Change Email" are shown. Below that, "Password: *****" and "Change Password" are shown. The bottom navigation bar is black with three icons: a triangle, a circle, and a square.

Change Email Address:

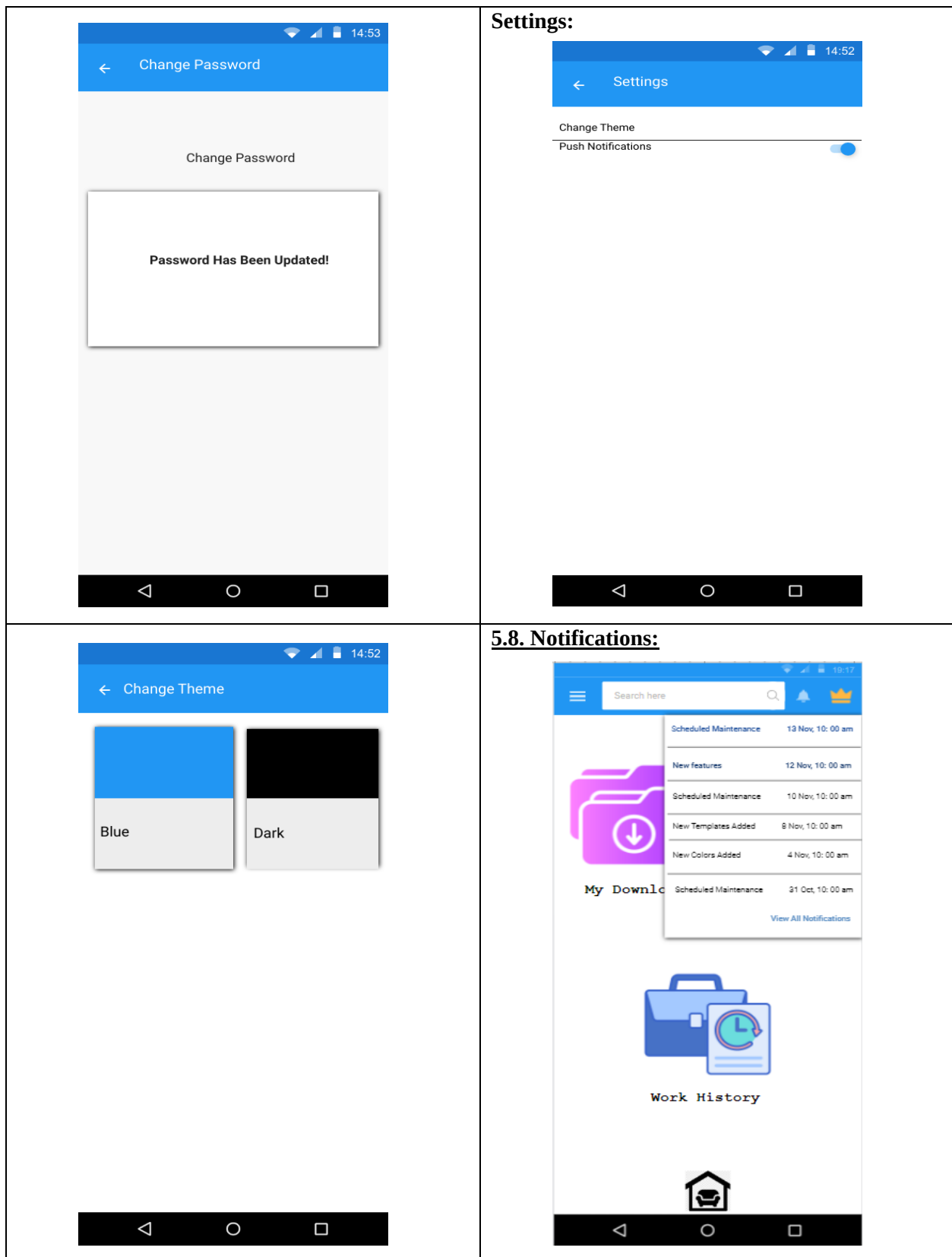
Mobile app change email screen. The header is blue with a back arrow and "Change Email". The status bar shows 14:52. Below the header, the title "Change Email" is centered. There are two input fields: "Current Email" and "New Email". Below them is a blue "Confirm" button. A keyboard is visible at the bottom, showing the letters "q w e r t y u i o p" on the first row, "a s d f g h j k l" on the second row, and "z x c v b n m" on the third row. The bottom navigation bar is black with three icons: a triangle, a circle, and a square.

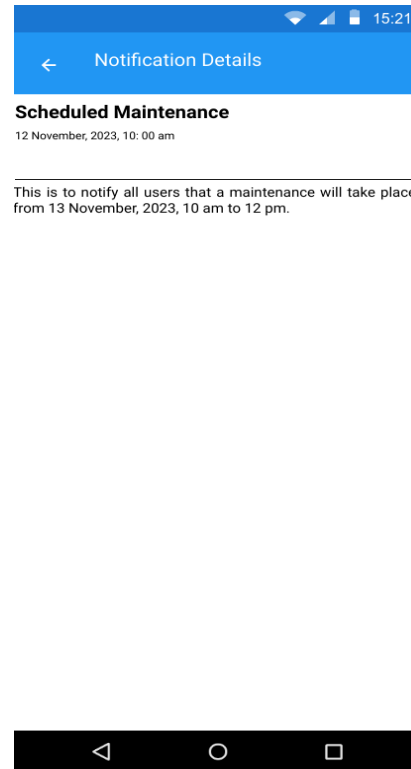
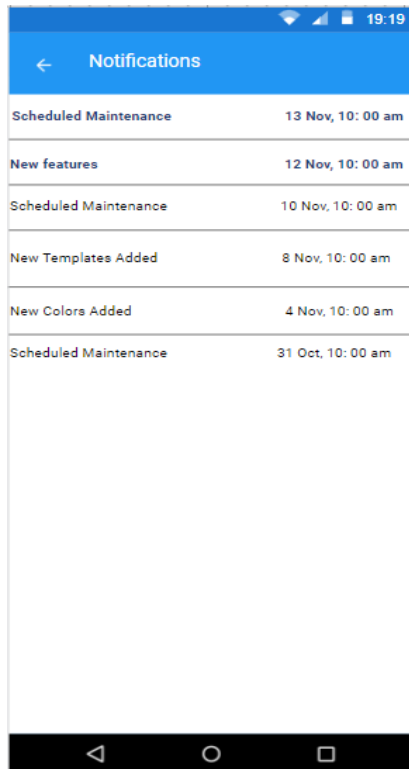


Mobile app change email confirmation screen. The header is blue with a back arrow and "Change Email". The status bar shows 14:52. Below the header, the title "Change Email" is centered. A white box with a grey border contains the text "Email Has Been Updated!". The bottom navigation bar is black with three icons: a triangle, a circle, and a square.

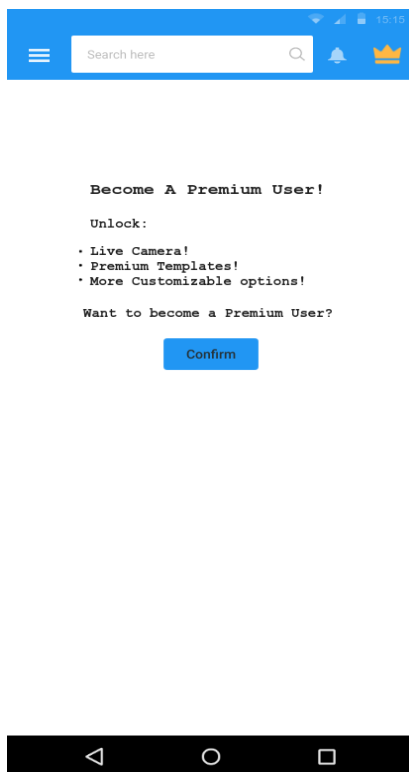
Change Password:

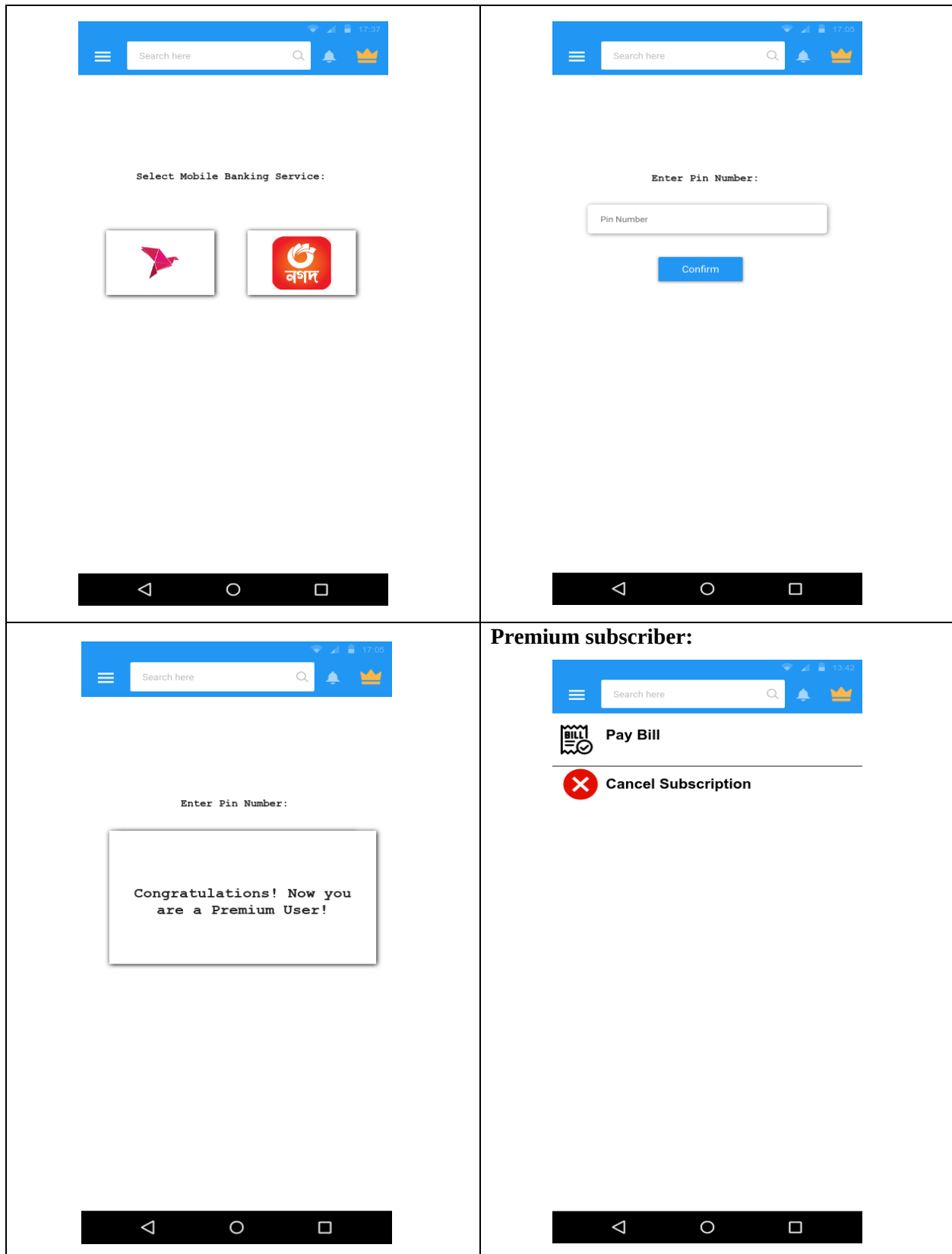
Mobile app change password screen. The header is blue with a back arrow and "Change Password". The status bar shows 19:02. Below the header, the title "Change Password" is centered. There are three input fields: "Current Password", "New Password", and "Confirm New Password". Each field has an eye icon to the right. Below them is a blue "Confirm" button. A keyboard is visible at the bottom, showing the letters "q w e r t y u i o p" on the first row, "a s d f g h j k l" on the second row, and "z x c v b n m" on the third row. The bottom navigation bar is black with three icons: a triangle, a circle, and a square.

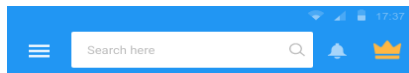




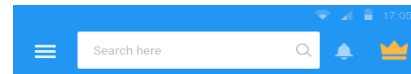
5.9. Subscription:





Pay Bill:


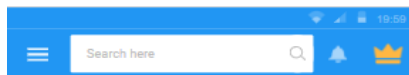
Select Mobile Banking Service:

Enter Pin Number:

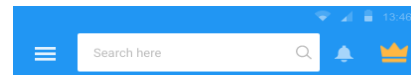
Pin Number

Confirm

Enter Pin Number:

Bill payment successful!


Pay Bill



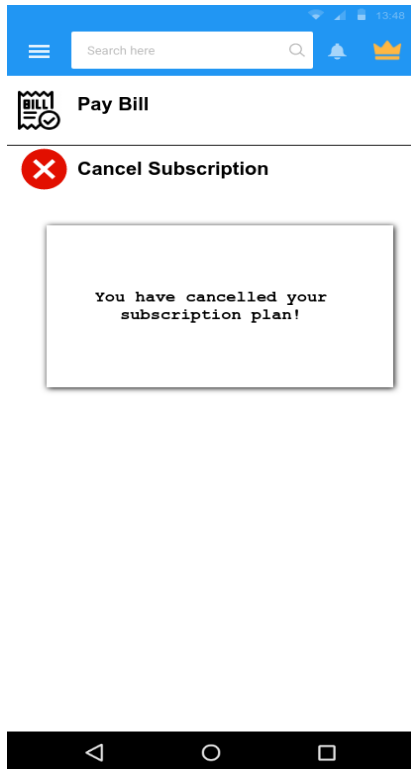
Cancel Subscription

Are you sure you want to cancel your subscription plan?

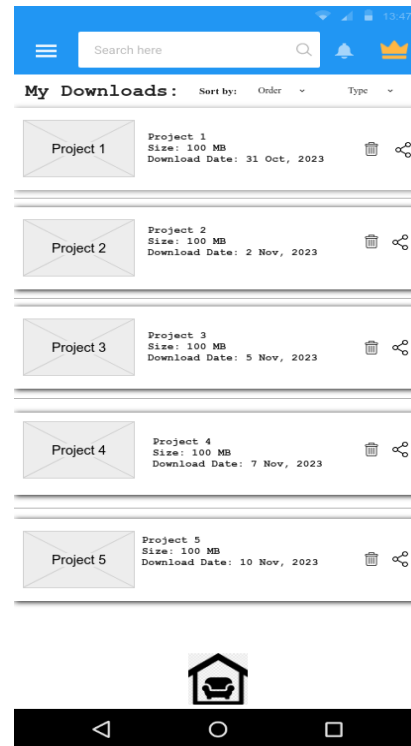
Yes

No

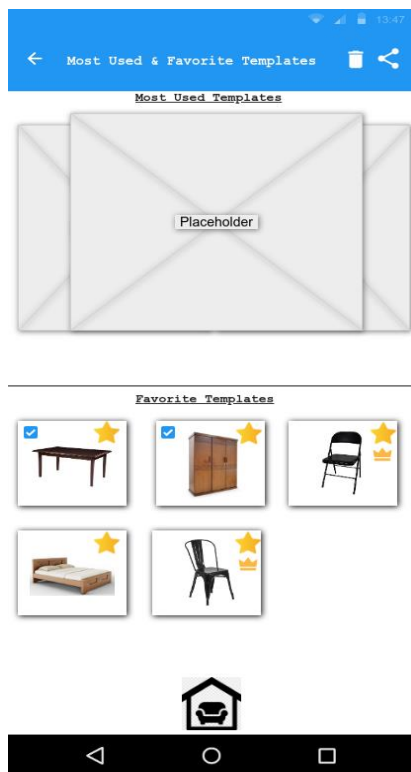




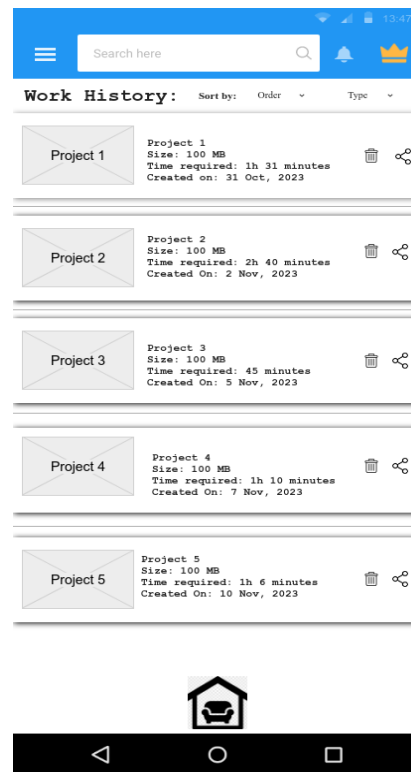
5.10. My Downloads:



5.11. Most used & Favorite Templates:



5.12. Work History:



6. PROJECT TEST PLANNING:

Project Name: Augmented Reality Software for Interior Design			Test Designed by: Shakil Jamil Mrida	
Test Case ID: FR_1			Test Design Date: 28/11/2023	
Test Priority: High			Test Executed by: Shakil Jamil Mrida	
Module Name: Login			Test Execution Date: 28/11/2023	
Test Title: Verify Login using username/email and password				
Description: Test app login				
Precondition (if any): User has an account.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open app. 2. Click “Sign In” 3. Enter username/email. 4. Enter password. 5. Click “Sign In”	Username: Shakil Jamil Mrida Password: 1NoThiNg!2	User will be logged in to the app.	As expected.	Pass.
Post Condition: User is validated with database and successfully login to account. The account session details are logged in the database.				

Project Name: Augmented Reality Software for Interior Design	Test Designed by: Shakil Jamil Mrida
Test Case ID: FR_2	Test Design Date: 28/11/2023
Test Priority: High	Test Executed by: Shakil Jamil Mrida
Module Name: SignUp session	Test Execution Date: 28/11/2023
Test Title: Verify registration of new user	
Description: Test app registration	
Precondition (if any): User has a valid email address	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open app. 2. Click “Sign Up”. 3. Enter Full name. 4. Enter DOB. 5. Enter valid mobile no. 6. Enter a valid email address. 7. Select Gender. 8. Enter password. 9. Confirm password. 10. Click “Agree to Terms & Conditions”.	Full Name: Shakil Jamil Mrida DOB: xx/xx/2000 Mobile no.: 018xxxxxxxx Email: sjm@gmail.com Gender: Male Password: 1NoThiNg!2	User should be registered to the database of the app.	As expected.	Pass
Post Condition: The account info of the user is stored in the database of the app.				

Project Name: Augmented Reality Software for Interior Design			Test Designed by: Shakil Jamil Mrida	
Test Case ID: FR_3			Test Design Date: 28/11/2023	
Test Priority: High			Test Executed by: Shakil Jamil Mrida	
Module Name: Forgot Password Session			Test Execution Date: 28/11/2023	
Test Title: Verify “Forgot Password?” function.				
Description: Test Forgot Password option.				
Precondition (if any): User has a valid account and access to their registered email.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Open app. 2. Click “Sign In”. 3. Click “Forgot Password?” 4. Enter Email Address. 5. Click By Email/By SMS. 6. If by email, click link sent to email.’ 7. If by SMS, type OTP. 8. Enter New Password. 9. Confirm New Password.	Email: sjm@gmail.com New Password: 2NoThiNg!1	User sets new password and can login again.	As expected.	Pass.
Post Condition: The new password of the user has been updated in the database and user can login with new password now.				

Project Name: Augmented Reality Software for Interior Design			Test Designed by: Md. Amir Hossain Alif		
Test Case ID: FR_4			Test Design Date: 01/12/2023		
Test Priority: High			Test Executed by: Md. Amir Hossain Alif		
Module Name: Search Bar Option.			Test Execution Date: 01/12/2023		
Test Title: Verify the search bar in the app.					
Description: Test the search bar function.					
Precondition (if any): User has logged in to the app.					
Test Steps		Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app. 2. Login to the app. 3. Enter search query.		Email: ah@gmail.com	The app will show templates	As expected.	Pass.

4. Select template/item. 5. Select or reset filters. 6. Enter sort. 7. Select or remove favorite.	Password: 999OP123 Search: Chair Filters: Black, Metal Order by: Popularity.	of black metal chairs to the user.		
Post Condition: If a user selects any template as their favorite, it will show in the favorites section and updated in the database.				

Project Name: Augmented Reality Software for Interior Design			Test Designed by: Md. Redowan Hossain	
Test Case ID: FR_5			Test Design Date: 30//11/2023	
Test Priority: High			Test Executed by: Md. Redowan Hossain	
Module Name: Design function.			Test Execution Date: 30//11/2023	
Test Title: Verify the various options and elements of the Design feature.				
Description: Test Design Feature				
Precondition (if any): User has logged in.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app. 2. Login to the app. 3. Select the “Design” icon. 4. Enter name of new project and select “Make New Project”. 5. Select either Camera Scan or Manual Modification. 6. Grant access to camera. 7. Take or upload photo. 8. Add template.	Email: rh@gmail.com Password: 623UZ6A52 Project Name: P_1 Mode: Camera Scan Take Photo Add Bed template and	Project is downloaded in “MyDownloads” and saved in Work History as well.	As expected.	Pass.

9. Resize template. 10. Edit. 11. If manual modification mode, use drawing board. 12. If manual modification mode, custom templated can be uploaded. 13. If premium user, live camera can be used along with extra templates and edit options. 14. Click “Ok” to save to Work History. 15. Click “Download” icon to save to “MyDownloads” along with Work History. 16. Click cancel to remove everything.	resize: 10 feet x 8 feet x 3 feet. Edit Download			
Post Condition: The project is saved in the database as well.				

Project Name: Augmented Reality Software for Interior Design	Test Designed by: Md. Redowan Hossain
Test Case ID: FR_6	Test Design Date: 30/11/2023
Test Priority: Medium	Test Executed by: Md. Redowan Hossain
Module Name: Side Navigation feature.	Test Execution Date: 30/11/2023.
Test Title: Verify the change email and password features, Settings features and log out features of the side navigation bar.	
Description: Test the side navigation of the app.	
Precondition (if any): User has logged in.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app. 2. Login to the app. 3. Slide the side navigation. 4. Select Profile. 5. Select Change Email. 6. Enter current email and new email address. 7. Select confirm. 8. Login again and this time select change password in the profile. 9. Enter current and new passwords and confirm new password. 10. Select confirm. 11. Login again and this time select Settings. 12. Turn on/off push notification. 13. Select "Change Theme" and select either blue or dark. 14. Go back and select "Log Out".	Email: rh@gmail.com Password: 623UZ6A52 New Email: rh1@gmail.com New Password: 6548QDE9966 Turn on Push Notifications. Theme: Dark.	User has changed his email and password, tuned on push notifications and changed their application theme to dark. After that they logged out.	As expected.	Pass.

Post Condition: User’s email, password has been updated in the database, along with their theme of choice and the state of their push notifications.				
Project Name: Augmented Reality Software for Interior Design			Test Designed by: Md. Amir Hossain Alif	
Test Case ID: FR_7			Test Design Date: 01/12/2023	
Test Priority: Medium			Test Executed by: Md. Amir Hossain Alif	
Module Name: Notifications feature.			Test Execution Date: 01/12/2023	
Test Title: Verify the notifications feature of the app.				
Description: Test the notifications option.				
Precondition (if any): User is logged in and connected to the internet.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app. 2. Login to the app. 3. Select the notifications icon. 4. Select “View all notifications.” 5. View any notification.	Email: ah@gmail.com Password: 999OP123 Choose notification.	User is able to view and see the details of the notifications.	As expected.	Pass.
Post Condition: Users can their notifications.				

Project Name: Augmented Reality Software for Interior Design			Test Designed by: Raiyan Sayeed	
Test Case ID: FR_8			Test Design Date: 29/11/2023	
Test Priority: High			Test Executed by: Raiyan Sayeed	
Module Name: Subscription feature.			Test Execution Date: 29/11/2023	
Test Title: Verify the options in the subscription feature.				
Description: Test the subscription feature.				
Precondition (if any): User is logged in				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Open the app. 2. Login to the app. 3. Select “Premium” icon. 4. Select “Confirm”. 5. Select Subscription plan. 6. Select Mobile banking service. 7. Enter Pin Number. 8. Select “Confirm”. 9. Now if premium users select “Pay Bill”, they need to select the mobile banking service first and then enter their pin no. and confirm. 10. Select “Cancel Subscription” to cancel premium subscription.	Email: Password: Subscription plan: Monthly. Mobile Banking service: Bkash. Pin: xxxx	User becomes a premium user.	As expected.	Pass.
Post Condition: The status of a user is changed to “premium” in the database.				

Project Name: Augmented Reality Software for Interior Design	Test Designed by: Nowrin Tasnim
Test Case ID: FR_9	Test Design Date: 02/12/2023
Test Priority: High	Test Executed by: Nowrin Tasnim
Module Name: MyDownloads feature	Test Execution Date: 02/11/2023
Test Title: Verify the features of the MyDownloads option.	
Description: Test the MyDownloads feature.	

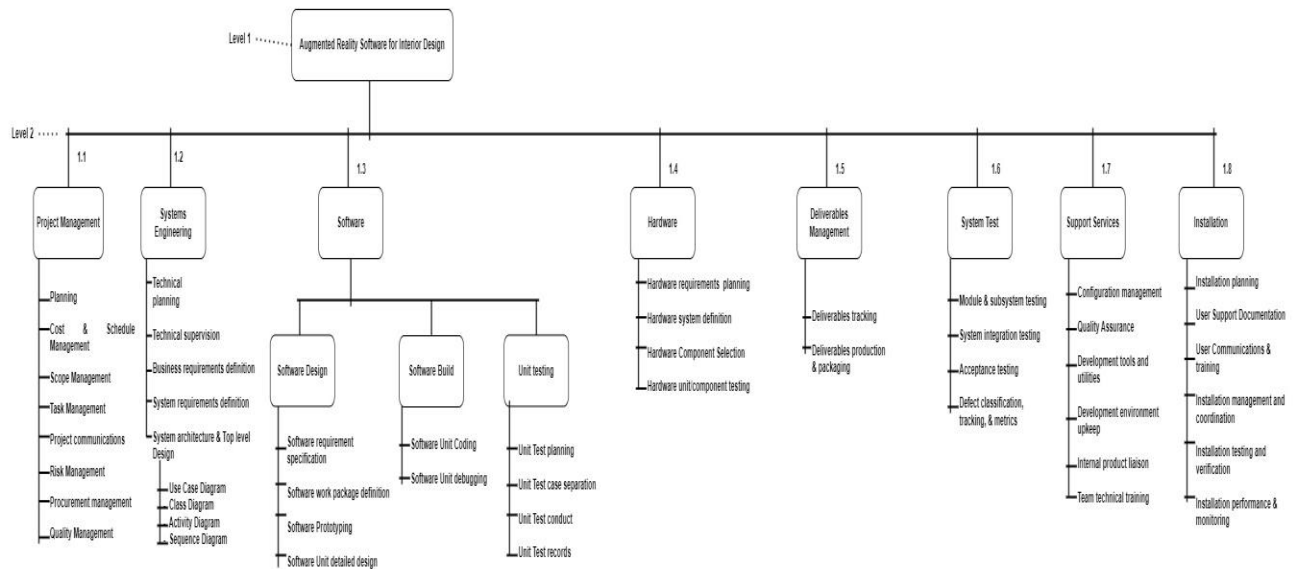
Precondition (if any): User is logged in.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app. 2. Login to the app. 3. Select “MyDownloads” icon. 4. Edit Project. 5. Select “Delete” option to delete project. 6. Select “Share” option to share the project.	Email: nt@gmail.com Password: 773CRT962 Select “Delete”	Project is deleted.	As expected.	Pass.
Post Condition: Project is deleted from MyDownloads if user selects the delete option.				

Project Name: Augmented Reality Software for Interior Design		Test Designed by: Nowrin Tasnim		
Test Case ID: FR_10.		Test Design Date: 02/12/2023		
Test Priority: High		Test Executed by: Nowrin Tasnim		
Module Name: Most used & Favorite Templates		Test Execution Date: 02/12/2023		
Test Title: Verify the feature in the Most used & Favorite Templates.				
Description: Test Most used & Favorite Templates features.				
Precondition (if any): User is logged in.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app. 2. Login to the app. 3. Select “Most used & Favorite Templates” icon. 4. Remove from Most Favorites.	Email: nt@gmail.com Password: 773CRT962 Unselect Metal chair from favorites.	Metal chair unselected from most favorites.	As expected.	Pass.

Post Condition: Templated is removed from the most favorites section of the user.

Project Name: Augmented Reality Software for Interior Design.		Test Designed by: Nowrin Tasnim		
Test Case ID: FR_11		Test Design Date: 02/11/2023		
Test Priority: High		Test Executed by: Nowrin Tasnim		
Module Name: Work History		Test Execution Date: 02/11/2023		
Test Title: Verify the features of the work history option.				
Description: Test Work history features.				
Precondition (if any): User is logged in.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app. 2. Login to the app. 3. Select “Work History” icon. 4. Edit Project. 5. Select “Delete” option to delete project. 6. Select “Share” option to share the project	Email: nt@gmail.com Password: 773CRT962 Select “Delete”	Project is deleted.	As expected.	Pass.
Post Condition: Project is deleted from MyDownloads if user selects the delete option.				

7. WORK BREAKDOWN STRUCTURE (WBS):



CoCoMo calculations:

Let us consider our project to be organic.

So, Coefficient<Effort Factor> = 2.4

Project complexity, P= 1.05

SLOC-dependent co-efficient, T = 0.38

So, Effort = PM = $2.4 * (5500/1000)^{1.05} = 14.374 \approx 14$

DM = $2.50 * (PM)^{0.38} = 6.884 \approx 7$

ST = PM/DM = $2.088 \approx 2$

8. TIMELINE CHART:

Total no. of weeks = $DM * 4 = 6.884 * 4 \approx 7 * 4 = 28$ weeks.

As we used SCRUM process model, the timeline chart for our project will be the following:

	Pre-Game			Development (Game) Phase																						Post-Game		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Weeks																												
Task/Person																												
A: Person 1 & Person 2																												
B: Person 2																												
C: Person 1																												
D: Person 2																												
E: Person 1																												
F: Person 2																												
G: Person 1																												
H: Person 2																												
I: Person 1																												
J: Person 2																												
K: Person 1																												
L: Person 2																												
M: Person 1																												
N: Person 2																												
O: Person 2																												
P: Person 1																												
Q: Person 2																												
R: Person 2																												
S: Person 1																												
T: Person 2																												
U: Person 1																												
V: Person 1 & Person 2																												
W: Person 1 & Person 2																												

Activity Keys:

A: Planning

B: Architecture

C: Module 1 Analysis & Design

D: Module 1 Coding

E: Module 1 Testing

F: Module 2 Analysis & Design

G: Module 2 Coding

H: Module 2 Testing

I: Module 3 Analysis & Design

H: Module 3 Coding

K: Module 3 Testing

L: Module 4 Analysis & Design

M: Module 4 Coding

N: Module 4 Testing

O: Module 5 Analysis & Design

P: Module 5 Coding

Q: Module 5 Testing

R: Module 6 Analysis & Design

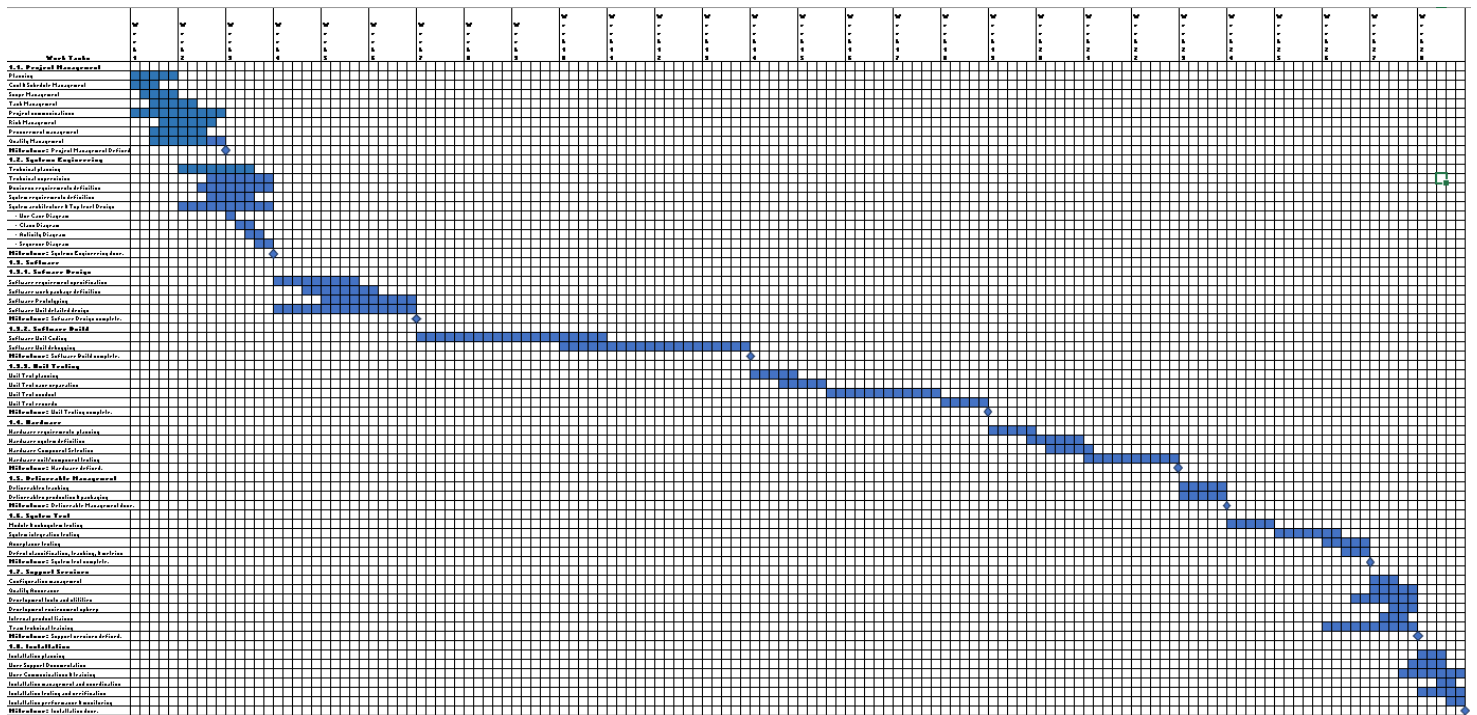
S: Module 6 Coding

T: Module 6 Testing

U: Integration Testing

V: System Testing

W: Documentation



EVA calculations:

Tasks	Planned Effort		Actual Effort	
1	↑	5	↑	7
2		3		3
3		4		3
4		5		4
5	BCWP	10		14
6		6	BCWS	5
7		6		5
8		8		7
9		8		10
10		7		7
11		8		...
12		5		...

Given, the total number of tasks = 48.

Effort estimated, BAC = PM x 22 = 14 x 22 = 308

BCWS = 5+3+4+5+10+6+6+8+8+7+8+5 = 75

BCWP = 5+3+4+5+10+6+6+8+8+7 = 62

So, SPI = BCWP/BCWS = 0.827

SV = BCWP – BCWS = 62 – 75 = -13 person-day

ACWP = 7+3+3+4+14+5+5+7+10+7 = 65

CPI = BCWP/ACWP = 62/65 = 0.954

CV = BCWP – ACWP = 62-65 = -3 person-day

% scheduled for completion = BCWS/BAC = 75/308 x 100% = 24.351 %

% completed = BCWP/BAC = 62/308 x 100% = 20.130 %

8. RISK ANALYSIS:

Risks	Category	Probability	Impact
Size estimate may be significantly low	PS	60%	2
Larger number of users than planned	PS	30%	3
Less reuse than planned	PS	70%	2
End users resist the system	BU	40%	3
Delivery deadline will be tightened	BU	50%	2
Funding will be lost	CU	40%	1
Customer will change requirements	PS	80%	2
Technology will not meet expectations	TE	30%	1
Lack of Experience with Tools	DE	80%	3
Staff inexperienced	ST	30%	3
Staff turnover will be high	ST	60%	2
Lack of communication between Team members	ST	60%	2
Redundant or excess functions	TE	50%	2
Inadequate infrastructure to support AR technology	DE	40%	3
Insufficient Project Documentation	PR	40%	2
Complex Infrastructure of Software given by Customer.	CU	60%	2

1 = catastrophic

2= critical

3= marginal

4= negligible

Rubric for Project Assessment (CO1)

Marking Criteria	Marks Distribution (Maximum 3X5=15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Background Analysis	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Analysis the impact of societal, health, safety, legal and cultural issues	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Existing Studies and Relevant Example	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					

Rubric for Project Assessment (CO2)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Argumentation of Model selection with Evidence of Argumentation	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Submission, Completeness, Spelling, grammar and Organization of the Project report	Project report is not complete and Several errors in spelling and grammar. Present a Confusing organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	Some errors in spelling and grammar. Some problems of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	Few errors in spelling and grammar. Presents most of the details in a logical flow of organization in definition, details, and example.	Project report is complete and No errors in spelling and grammar. Consistently presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					

CO Pass / Fail:	
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